

Should banks' regulatory capital reflect unrealized capital gains and losses? A quantitative assessment*

Vedant Agarwal[†] Jan Schäfer[‡]

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Abstract

We develop a quantitative general equilibrium model of risky banks with a bond portfolio subject to interest rate risk and consider the implications of having the unrealized capital gains or losses of such a portfolio excluded from the (amortized-cost accounting) or included in the (fair-value accounting) definition of regulatory capital. We show that unrealized gains or losses affect bank lending despite the deposit franchise. Banks are relatively insulated from short-term volatility in securities returns under the amortized-cost accounting treatment, resulting in a smoother credit supply. This comes at the cost of higher bank failure probabilities during prolonged periods of intense monetary policy tightening. Under our calibration, fair-value accounting treatment is superior in welfare terms.

Keywords: Interest rate risk, credit risk, regulatory capital, accounting rules, financial stability

JEL Classification: G21, G28, G32, G33, E44, E51

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[†]CAFRAL: vedant.agarwal@cafral.org.in

[‡]CEMFI: jan.schaefer@cemfi.edu.es

1 Introduction

Banks invest a considerable fraction of their asset holdings in long-term fixed-income securities which are highly liquid and nearly free of default risk. These securities allow the bank to meet any unexpected withdrawals of its predominantly short-term funding, while simultaneously earning a term premium. But they also pose a risk to the bank because the market value of long-term securities declines when the yield curve shifts upwards, producing accounting losses that deplete their equity capital. Should such accounting losses (or gains) related to the impact of interest rate risk on debt securities also affect banks' regulatory capital base? Basel III contains a proposal in favor of it. Yet, some bank managers, regulators and academics have argued that doing so would result in greater volatility in banks' regulatory capital levels (see e.g. [Barth, Landsman, and Wahlen, 1995](#); [Basel Committee on Banking Supervision, 2017](#)).¹ Indeed if banks are constrained in their ability to raise equity, then shielding their regulatory capital from some of the valuation implications of temporary fluctuations in interest rates can allow them to maintain a more steady provision of credit. However, during a prolonged period of unexpectedly high interest rates, neglecting unrealized losses in the portfolio of debt securities may allow banks to operate with a severely inflated regulatory equity valuation and consequently high effective leverage, posing a financial stability risk (see e.g. [Flannery and Sorescu, 2023](#)). Arguments supporting this view are reflected in the Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank ([Barr, 2023](#)).

This paper studies this trade-off in a quantitative macro-banking model that jointly accounts for banks' exposure to credit and interest rate risk, and in which financial stability risks from unexpectedly high interest rates are most severe when accompanied by high default rates in banks' loan portfolio. In the model, banks use insured deposits and equity to extend one-period risky loans to entrepreneurial firms and invest in a portfolio of default-free long-term government bonds. Firm defaults cause losses for banks, depleting their equity capital. The market price of long-term bonds is sensitive to changes in the monetary policy rate set by the central bank. In the baseline, the resulting unrealized (accounting) gains or losses are excluded for calculating the value of bank equity relevant for regulatory requirements – which as in reality applies a positive risk weight to the risky loans and a zero weight

¹Regulators have long applied so-called prudential filters to (fully or partially) shield regulatory equity from fluctuations in the market value of debt securities. See [Argimón, Dietsch, and Estrada \(2018\)](#) for a discussion on the pre-crisis heterogeneity across Euro Area countries in their use of the prudential filters. [Kim, Kim, and Ryan \(2019\)](#) provide a concise overview on the accounting and capital requirements for securities held by U.S. banks.

on the long-term government bonds. Banks' regulatory capital ratio therefore overstates the actual value of their equity during periods of increasing interest rates. Distortions due to limited liability combined with safety net guarantees imply that banks operate with particularly high effective leverage during such periods, so that the impact of credit losses on their solvency and the macro-economy are much larger.

The calibrated macro-banking model of credit and interest rate risk reproduces many relevant features of the data related to financial markets, macro-economic quantities and prices. We also validate the model in its ability to capture the differences coming from the regulatory treatment of accounting losses by comparing it with the empirical literature on the effects that realized and unrealized capital losses have on loan pricing by banks. Consistent with available evidence, the model implies that banks charge a higher lending rate in response to unrealized balance sheet losses, and that this response is weaker than the one associated with realized changes in regulatory capital.

We then compare the baseline regulatory accounting regime to an alternative regime that includes unrealized gains and losses on the bond portfolio in calculating the value of bank equity relevant for regulatory purposes. Not surprisingly, regulatory bank equity is more sensitive to changes in the interest rate, which translates into a greater volatility in credit. At the same time, the terms of the financial contract between banks and firms are better aligned with macro-economic conditions and bank balance sheet fundamentals. The better pricing of risk makes both banks and firms safer on average, freeing up resources otherwise spent on deadweight losses from bankruptcies, resulting in a higher level and lower volatility of consumption. Overall, this regulatory regime is therefore welfare-improving, lending support to the Basel III proposal for removing prudential filters which insulate banks' regulatory capital from accounting gains or losses on debt securities portfolio.

Our model embeds a credit-risk framework à la Mendicino et al. (forthcoming) in a standard New-Keynesian model with investment. The credit-risk framework adds banks in a setup following the [Bernanke, Gertler, and Gilchrist \(1999\)](#) financial-accelerator tradition and explicitly takes into account the structure of asset returns implied by holding of risky loans whose risk of default is not fully diversifiable at the bank level. As a result, bank solvency problems arise endogenously from high default rates among bank borrowers. Finally, we augment the bank's portfolio choice problem to include default-free long-term bonds issued by the government, with the maturity mismatch resulting from holding these bonds exposing the bank to interest rate risk.

Banks run a deposit franchise, setting deposit rates that are a fixed fraction of the policy

rate. Their deposit spreads are therefore increasing in the policy rate (see [Drechsler, Savov, and Schnabl, 2021](#)). This partially hedges interest rate risk. In our model, the deposit franchise has an additional effect which is that it makes the bank’s objective function convex in lending. This is because payoffs from the deposit franchise are riskless while lending is risky. Hence, banks are safe when they supply loan volumes up to the level at which the highest possible loss in their loan portfolio equals the profits from the deposit franchise. For higher levels of lending, banks have a non-zero default probability that is increasing in the size of their loan portfolio. Limited liability and safety net guarantees imply that once banks are risky, their marginal profit from lending must change (since at that point, any increase in lending affects the bank’s default probability). For a given loan rate, banks then either prefer to lend the maximum amount that respects the regulatory capital requirement, or zero. In equilibrium, bank equity is sufficiently scarce, such that the loan rate clearing the market is high enough for all banks to lend up to the maximum amount consistent with the regulatory capital requirement.

We estimate the model parameters using the simulated method of moments, targeting a large set of unconditional moments in macro, banking, and financial euro area (EA) data from the OECD Quarterly National Accounts and the ECB Statistical Data Warehouse for 1995-2016. Over this period, banks’ loans-to-bonds ratio averaged 3.6 and the maturity of their bond holdings corresponded to about 3.4 years. Our baseline calibration replicates these features of the data, which are crucial for capturing the exposure to interest rate risk coming from these assets. The capital requirement on the risky loans is set at 8%, matching the standards set by the Basel agreements.

Our main quantitative exercises compare the baseline regulatory accounting regime in which the value of bank equity relevant for regulation reflects the value of the long-term bonds at amortized-cost (or “book value”) with a regime in which the value of equity reflects the fair value (or “market value”) of the bonds. We validate the model’s ability to allow this comparison by assessing it against the empirical literature on loan pricing implications of realized and unrealized bank losses. Consistent with available evidence, our model implies that banks charge a higher lending rate in response to unrealized losses on their balance sheet ([Volk, 2024](#)), and that this response is weaker than the one associated with realized changes in regulatory capital ([Dagher, Dell’Ariccia, Laeven, Ratnovski, and Tong, 2016](#)). Furthermore, the model generates episodes of large unrealized losses that are broadly consistent with the historical experience during the post-COVID tightening cycle, when inflationary pressures and the ensuing aggressive monetary policy hikes led to substantial valuation losses on banks’ long-term bond portfolios, as documented by [Marsh and Laliberte \(2023\)](#).

The validated model is used to study the optimal prudential measure of regulatory bank capital. When unrealized losses on the long-term bonds are excluded, banks operate with effectively high leverage during periods of monetary policy tightening. This increases the risk of bank insolvencies due to borrower defaults, because the loss-absorbing capacity of a bank’s regulatory equity is lower once borrower defaults are sufficiently high to force the bank to realize losses in the bond portfolio. But banks under-price borrower risk, both due to the presence of safety net guarantees and limited liability, and because they do not internalize the effects of their individual choices on the social costs of their failures. Banks extend loans up to the point that the capital requirement is binding – which in this case is less restrictive than if it were to be based on their true loss-absorbing capacity. While this dampens the decline in credit, investment and output, any adverse shock that increases default risk among bank borrowers causes a large increase in bank defaults.

Instead, when unrealized gains or losses on the long-term bonds are included, regulatory bank capital is more sensitive to changes in interest rates. As a result, credit is more volatile. However, firm and bank default probabilities are *less* volatile. This is because this regulatory treatment limits banks’ effective leverage (and hence riskiness) in times when it would otherwise be elevated due to unrealized losses and thus makes the banking sector more resilient to credit losses in times when it would be particularly vulnerable to them. Hence, it reduces the mean and variance of bank and firm failures. The associated savings in deadweight loss costs of default lead to a higher level and lower volatility of consumption, translating to higher welfare. Our results thus lend support to the Basel III proposal of recognizing unrealized gains and losses for regulatory purposes (i.e., removing prudential filters).

Related literature This paper belongs to the literature on quantitative models of bank regulation. Following the formalization of welfare trade-offs associated with capital regulation in [Van den Heuvel \(2008\)](#), a big strand of the literature has studied the optimal level of capital requirement ([Begenau, 2020](#); [Elenev, Landvoigt, and Van Nieuwerburgh, 2021](#); [Abad, Martínez-Miera, and Suarez, 2024](#); [Mendicino et al., forthcoming](#)), the interaction of capital regulation and monetary policy ([Angeloni and Faia, 2013](#); [Collard, Dellas, Diba, and Loisel, 2017](#)), the interaction of capital regulation and banking market structure ([Corbae and D’Erasmus, 2021](#); [Begenau and Landvoigt, 2022](#)), and the rationale behind dynamic capital requirements ([Davydiuk, 2017](#); [Malherbe, 2020](#)). In the models considered in this literature, banks’ assets and liabilities are typically assumed to last only one period, thus rendering the discussion on interest rate risk and the capital treatment of unrealized gains or losses irrele-

vant. We extend the analysis adding long-term bonds to banks’ assets and complement the existing literature by focusing on the implications of the prudential treatment of unrealized gains and losses for credit supply and banks’ solvency.

Our work is more closely related to [Begenau, Bigio, Majerovitz, and Vieyra \(2025\)](#), in which differences in banks’ book equity (relevant for regulation) and market equity arise from accounting rules which allow a delayed recognition of loan losses. As a result, regulation requiring instantaneous recognition of loan losses effectively tightens capital requirements. In contrast, the regulatory requirement to recognize revaluations of the long-term bonds portfolio in our model results in tighter capital requirements during monetary policy tightening and weaker requirements during monetary policy expansions.

Our work is related to the literature on “accounting” (or capital measurement) issues in banking.² Several studies in this literature share a common theme that fair value accounting may lead financial institutions subject to capital regulation to conduct fire sales of illiquid or credit-risky assets (carrying a positive regulatory risk-weight, such that selling the asset improves the regulatory-equity-to-risk-weighted-assets ratio).³ In the wake of the 2023 U.S. regional banking crisis involving the failure of Silicon Valley Bank, these accounting discussions have focused on the prudential treatment of unrealized gains or losses on liquid debt securities that are relatively free of default risk, such as US government bonds (which carry a zero regulatory risk weight). [Greenwald, Krainer, and Paul \(2024\)](#) and [Orame, Ramcharan, and Robatto \(2025\)](#) study how the regulatory accounting framework influences the transmission of monetary policy onto bank lending. The implications produced by our model regarding the same are consistent with these and other empirical findings (see e.g. [Beutler, Bichsel, Bruhin, and Danton, 2020](#); [Marsh and Laliberte, 2023](#)). We complement these studies by underscoring the financial stability consequences of the regulatory accounting framework, so our work bridges this literature with the renewed discussions on the importance and implications of banks’ interest rate risk exposure ([Drechsler et al., 2021](#); [Drechsler, Savov, Schnabl, and Wang, 2023](#); [Haddad, Hartman-Glaser, and Muir, 2023](#); [Jiang, Matvos, Piskorski, and Seru, 2024](#); [DeMarzo, Krishnamurthy, and Nagel, 2024](#); [Begenau, Landvoigt, and Elenev, 2024](#); [Varraso, 2024](#)).

²See [Freixas and Tsomocos \(2004\)](#) for an early theoretical contribution.

³For example, [Allen and Carletti \(2008\)](#) show that when financial markets are illiquid, such as during a financial crisis, the use of fair value accounting to assess solvency may lead to undesirable liquidation of banks’ assets. See also [Plantin, Sapra, and Shin \(2008\)](#). [Ellul, Jotikasthira, Lundblad, and Wang \(2015\)](#) argue that when regulatory capital reflects fair value of risky assets, it creates incentives for financial institutions to sell risky assets during a financial crisis since it improves their risk-weighted capital ratios. However, the evidence presented in [Laux and Leuz \(2010\)](#) suggests it is unlikely that fair value accounting contributed to the severity of the 2008 financial crisis in a sizable way.

Outline The rest of the paper is organized as follows. Section 2 describes our macro-banking model. Section 3 contains the solution method and calibration strategy, presents the baseline parameterization, and discusses the quantitative performance of the model. In Section 4 we analyze the performance of the economy under alternative regulatory accounting frameworks, identifying the approach that maximizes social welfare. The Appendix contains a complete list of equilibrium conditions and full description of the data sources, solution method, and several complementary materials referred throughout the main text.

2 The Model

We consider a discrete-time, infinite-horizon economy in which dates are indexed by t . The baseline framework is a financial accelerator model embedded in a standard New Keynesian model with investment.⁴ Following the tradition of [Gertler and Karadi \(2011\)](#) each household consists of workers, entrepreneurs, and bankers. Workers supply labor to the production sector and transfer their wage income back to the household. Entrepreneurs and bankers provide equity to entrepreneurial firms and banks, respectively.

There exist a continuum of measure one of islands. In each island there is a continuum of measure one of entrepreneurial firms and a representative bank. Entrepreneurial firms and banks live for one period, issue equities among entrepreneurs and bankers, respectively, and obtain external financing by issuing non-contingent debt in the form of bank loans and deposits, respectively. Entrepreneurial firms use equity and loans to buy physical capital, which some intermediate good producers rent in the next period. Their terminal net worth is subject to both idiosyncratic and island-specific shocks. The latter is non-diversifiable from the banks' perspective. In addition to providing loans, banks invest in a portfolio of long-term bonds. Both entrepreneurial firms and banks operate under limited liability and default when their terminal asset value is lower than their debt obligations. Non-defaulted entrepreneurial firms and banks pay their terminal net worth to entrepreneurs and bankers, respectively.

In the rest of this section, we present the model ingredients in more detail.

⁴See e.g., [Smets and Wouters \(2007\)](#) and [Christiano, Eichenbaum, and Evans \(2005\)](#).

2.1 Households

There is a unit continuum of households indexed by h , that provide consumption insurance to three types of members: workers, bankers and entrepreneurs.

Households derive utility from consumption C_t and disutility from labor H_t . To improve the quantitative performance of the model, consumption is subject to internal habit formation governed by the parameter b .⁵ Households provide differentiated labor hours H_{ht} to intermediate goods-producing firms, remunerated at a nominal wage W_{ht} . The disutility derived from labor is governed by the inverse Frisch elasticity φ_H and a scaling parameter ξ_H .

Households can save in fully insured bank deposits D_t remunerated at gross interest rate R_{Dt} . To account for non-bank funding, households can also invest in physical capital K_t^H at real price Q_t , subject to a management cost ς_t , and rent it to intermediate good producers at rate z_t .⁶ Physical capital depreciates at rate δ . The nominal price of the single consumption good is denoted by P_t , and inflation is defined as $\Pi_t = P_t/P_{t-1}$ with steady state $\bar{\Pi}$.

With all these ingredients, the maximization problem of household h is stated as:

$$\max_{\{C_{ht}, D_{ht}, K_{ht}^H, H_{ht}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_{ht} - bC_{ht-1})^{1-\sigma}}{1-\sigma} - \frac{\xi_H H_{ht}^{1+\varphi_H}}{1+\varphi_H} \right], \quad (1)$$

subject to the budget constraint:

$$\begin{aligned} P_t C_{ht} + P_t D_{ht} + P_t(Q_t + \varsigma_t) K_{ht}^H &= W_{ht} H_{ht} \\ + R_{Dt-1} P_{t-1} D_{ht-1} + P_t[z_t + (1-\delta)Q_t] K_{ht-1}^H &+ \Sigma_{ht}, \end{aligned} \quad (2)$$

where Σ_{ht} summarizes other cash flows that the household receives, but which are irrelevant for its optimization problem. We assume that the household invests its deposits symmetrically in all the (symmetric) banks in the economy. Appendix A.1 provides the FOCs for this problem.

⁵See, for example, [Smets and Wouters \(2007\)](#), [Christiano et al. \(2005\)](#), and [Christiano, Motto, and Rostagno \(2014\)](#).

⁶Capturing the non-bank-dependent part of the economy prevents our model from overstating the macroeconomic consequences of changes in credit supply.

2.1.1 Nominal Wage Setting

The model features sticky wages.⁷ A labor union collects all household-differentiated varieties of labor H_{ht} , which are sold to a competitive labor packer after setting nominal wages W_{ht} . The elasticity of substitution between varieties is ϵ_W . Wage setting is subject to Rotemberg (1982) adjustment costs governed by parameter θ_W which the union finances by charging households a lump-sum fee. Since these elements are standard in New Keynesian models, further details are relegated to Appendix A.2.

2.1.2 Bankers & Entrepreneurs

Bankers and entrepreneurs are modeled in a symmetric manner, and are therefore discussed together in this section.

At date t , bankers and entrepreneurs invest symmetrically in an all-islands portfolio of one-period banks and entrepreneurial firms, respectively. Bankers and entrepreneurs receive the terminal net worth of their banks and firms at the beginning of $t + 1$. At that point, bankers are also charged a lump sum tax by the government to finance the deposit insurance agency (DIA).

To make bankers and entrepreneurs net worth scarce, we assume that in every period a fraction $(1 - \theta_\chi)$, $\chi \in \{B, E\}$ of bankers and entrepreneurs retire and become workers, while the same measure of workers becomes bankers and entrepreneurs. When they retire, they pay out their wealth to households. New bankers and entrepreneurs in period t on the other hand receive a fraction ξ_χ of the net worth of the bankers and entrepreneurs that have retired in period t . Calibration of the parameters will ensure equity is scarce enough for banks and entrepreneurs never to finance all their investments without debt.

In every period, continuing and new bankers and entrepreneurs decide how much (real) dividends ν_t^χ to pay out to their households and how much (real) equity χ_t (with $\chi_t = B_t$ for bankers, and $\chi_t = E_t$ for entrepreneurs) to invest in the equity portfolio. Bankers and entrepreneurs take the nominal return on their equity ρ_t^χ as given. Stating the maximization

⁷As discussed in, e.g., Galí (2015) and Smets and Wouters (2007), sticky wages contribute to dampen the rise in inflation after, e.g., an expansionary monetary policy shock, consistent with data. In our setup this adds realism to the impact of debt deflation on financial sector defaults.

problem in real terms, the value function of a representative banker or entrepreneur i is

$$V_t^x(\chi_{it}) = \max_{\nu_{it}^x \geq 0, \chi_{it} \geq 0} \mathbb{E}_t \left[\nu_{it}^x + \mathbb{E}_t \Lambda_{t,t+1} \left((1 - \theta_\chi) \chi_{t+1} + \theta_\chi V_{t+1}^x(\chi_{it+1}) \right) \right], \quad (3)$$

where $\Lambda_{t,t+1}$ is the household's real stochastic discount factor, and

$$\chi_{it+1} = \frac{\rho_{t+1}^x}{\Pi_{t+1}} (\chi_{it} - \nu_{it}^x). \quad (4)$$

Following the established approach in the literature, we guess and verify that the value function is linear in the net worth of banker or entrepreneur i : $V_t^x(\chi_{it}) = s_t^x \chi_{it}$.⁸ Further guessing (and later verifying) that in the vicinity of the steady state we have $s_t^x > 1$, which implies $\nu_t^x = 0$ by the Envelope Theorem.⁹ It then follows that

$$s_t^x = \mathbb{E}_t \underbrace{\Lambda_{t,t+1} (1 - \theta_\chi + \theta_\chi s_{t+1}^x)}_{=\Lambda_{t,t+1}^x} \frac{\rho_{t+1}^x}{\Pi_{t+1}}. \quad (5)$$

Equation (5) defines the bankers', or entrepreneurs', stochastic discount factor for later use as $\Lambda_{t,t+1}^x = \Lambda_{t,t+1} (1 - \theta_\chi + \theta_\chi s_{t+1}^x)$. Finally, the aggregate law of motion of equity of bankers or entrepreneurs is

$$\chi_{t+1} = [\theta_\chi + \xi_\chi (1 - \theta_\chi)] \frac{\rho_{t+1}^x}{\Pi_{t+1}} \chi_t - \frac{T_{t+1}}{P_{t+1}}, \quad (6)$$

where T_{t+1} are nominal lump-sum taxes imposed by the deposit insurance agency, described below in detail.

2.2 Entrepreneurial Firms

Entrepreneurial firms provide the key connection between the financial sector and the real economy: they rely on bank loans to invest in physical capital used in the production sector. They hence transmit conditions in the financial sector to the real economy through their (physical) capital supply, and in turn transmit conditions in the real economy to the financial sector through the impact of the real return on (physical) capital on the loan default probability.

⁸See, for example, [Gertler and Kiyotaki \(2010\)](#).

⁹We make sure that under our calibration of the model parameters $s_t^x = 1$ with a probability close to 0 and thus directly impose $\nu_t^x = 0$.

Each island is populated by a unit continuum of entrepreneurial firms indexed by j . These are one-period institutions owned by entrepreneurs. Entrepreneurial firms purchase physical capital K_{jt}^E from capital producers at real price Q_t . To finance their investment, they use loans L_{jt} from the bank on their island and equity E_{jt} :

$$E_{jt} + L_{jt} = Q_t K_{jt}^E. \quad (7)$$

At date $t + 1$, entrepreneurial firms rent capital acquired at the end of t to intermediate good producers against a rental price z_{t+1} , and sell undepreciated capital $(1 - \delta)K_{jt}$ back to capital producers at real price Q_{t+1} .¹⁰ Following Mendicino et al. (forthcoming), the final asset value of every entrepreneurial firm is subject to an idiosyncratic shock ω_j and an island specific shock ω_k , engendering loan default risk which is only partly diversifiable at the island-specific banks. By limited liability, the nominal terminal net worth of entrepreneurial firm j on island k at time $t + 1$ is

$$P_{t+1} \Omega_{jkt+1}^{Firm}(\omega_j, \omega_k) = \max\{\omega_j \omega_k [P_{t+1} Q_{t+1} (1 - \delta) K_{jt}^E + P_{t+1} z_{t+1} K_{jt}^E] - R_{Ljt} P_t L_{jt}, 0\}. \quad (8)$$

According to Equation (8), entrepreneurial firm j defaults at $t + 1$ if its idiosyncratic shock is below the threshold $\bar{\omega}_{Ft+1}(\omega_k)$:

$$\bar{\omega}_{Ft+1}(\omega_k) = \frac{R_{Ljt} L_{jt}}{\omega_k \Pi_{t+1} [Q_{t+1} (1 - \delta) K_{jt}^E + z_{t+1} K_{jt}^E]}. \quad (9)$$

To capture the impact of uncertainty on the fluctuation of default risk, we introduce Christiano et al. (2014) risk shocks in the same way as Mendicino et al. (forthcoming). Specifically, we assume the shocks ω_j and ω_k are independent and log-normal distributed, with time-varying mean and variance:

$$\ln(\omega_{\Xi}) \sim N\left(-\frac{\sigma_{\omega_{\Xi t}}^2}{2}, \sigma_{\omega_{\Xi t}}^2\right), \quad \Xi \in \{j, k\}, \quad (10)$$

where the standard deviation $\sigma_{\omega_{\Xi t}}$ follows:

$$\ln(\sigma_{\omega_{\Xi t}}) = (1 - \rho_{\omega_{\Xi}}) \ln(\bar{\sigma}_{\omega_{\Xi}}) + \rho_{\omega_{\Xi}} \ln(\sigma_{\omega_{\Xi t-1}}) + \sigma_{\sigma_{\Xi}} \epsilon_{\sigma_{\Xi}}, \quad \epsilon_{\sigma_{\Xi}} \sim N(0, 1). \quad (11)$$

¹⁰In contrast to models in which all production factors are pre-financed with loans (e.g., Mendicino et al. forthcoming and Hristov and Hülsewig, 2017), our setup with only pre-financed capital allows output at t to respond to the monetary policy rate, as in the canonical New Keynesian model.

While the risk shocks $\epsilon_{\sigma_{\Xi}}$ are mean preserving ($\mathbb{E}(\omega_{\Xi t}) = 1 \forall t$), a higher value of $\sigma_{\omega_{\Xi t}}^2$ implies that the distribution of ω 's has fatter tails, leading to higher default risk.

At the end of each period, all terminal net worth of entrepreneurial firms is paid out to entrepreneurs. The nominal return on entrepreneurial equity is:

$$\rho_{t+1}^E = \frac{\Pi_{t+1} \int_0^\infty \int_0^\infty \Omega_{jkt+1}^{Firm}(\omega_j, \omega_k) dF_{jt+1}(\omega_j) dF_{kt+1}(\omega_k)}{E_t}. \quad (12)$$

2.3 Banks

Each island is populated by a representative bank k . As entrepreneurial firms, banks are active between two consecutive periods t and $t + 1$. In period t banks combine equity B_{kt} from bankers and insured deposits D_{kt} from households in order to extend loans L_{kt} to entrepreneurial firms operating in their island. The bank can also invest in both one-period government bonds S_{kt} , remunerated at the policy rate R_t set by the central bank, and long-term zero-coupon bonds S_{kt}^L , trading at market price Q_t^S . The latter are purchased by banks from a *bond management company* in period t and, if not maturing, resold to it in period $t + 1$. The role of this company is discussed below in detail. For tractability, as in [Hatchondo and Martínez \(2009\)](#) and [Chatterjee and Eyigungor \(2012\)](#), these bonds are assumed to reach maturity in an independent random manner with probability $\frac{1}{m}$ per period so that in each period a fraction $\frac{1}{m}$ of them mature. This implies that the average maturity of the bonds is m periods. In market value terms, banks face the following balance-sheet constraint

$$L_{kt} + S_{kt} + Q_t^S S_{kt}^L = B_{kt} + D_{kt}. \quad (13)$$

Banks extend loans in a perfectly competitive manner. It is assumed that in addition to the loan rate R_{Lt} , banks reap a non-pecuniary benefit c_R per unit of lending, reflecting the value of lending relationships (recent empirical evidence of the value of such relationships to banks is provided in [Agarwal, Chomsisengphet, Liu, Song, and Souleles \(2018\)](#), with [Boot \(2000\)](#) surveying the theoretical underpinnings). It is thus costly for the bank to reduce lending beyond the bank's lending capacity implied by capital requirements (to be discussed below), reflecting a weakening in lending relationships.¹¹ The role of this assumption will be

¹¹While each bank is a one-period institution, the bank's shareholders continue to profit from the banking relationship in the future. While we do not model such dynamics, this can for example be thought of as relationships embodied in the employees of the bank k (documented in [D'Andrea and Stivella, 2025](#)), not in the institution per se, with a constant set of employees employed by the succession of banks on island k . Each bank is managed in the interest of shareholders, and internalizes the benefits of the lending relationship

discussed below, alongside the bank's profit maximization problem.

Banks raise insured deposits D_{kt} in a monopolistic manner at rate R_{Dkt} . The demand for deposits of bank k is:

$$D(R_{Dkt}, R_{Dt}) = \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t. \quad (14)$$

Banks take the aggregate deposit rate $R_{Dt} = (\int_0^1 R_{Dkt}^{1-\epsilon_D} dk)^{\frac{1}{1-\epsilon_D}}$ as given. As shall be seen, this allows our model to feature a sensitivity of deposit rates to the policy rate R_t of the same form as in Drechsler et al. (2023), with deposit spreads increasing in the policy rate. Further, in keeping with the Drechsler et al. (2023) deposit franchise setup, the bank has to pay a fixed cost (in real terms) c_f per period to operate. This feature has two roles. First, as discussed in Drechsler et al. (2023), it allows to think of the deposit franchise as an interest rate swap, with a fixed leg c_f and a floating leg corresponding to the profits in deposit taking. Second, it allows the model to reproduce a realistic return on equity for banks.

Lastly, at the beginning of period t , every bank k is endowed with an identical amount of equity $\bar{B}_t$ by bankers, such that $B_{kt} = \bar{B}_t$.

Intertemporal trade of the long-term bonds. The banking industry operates a long-term bond management company which centralizes the trade of bonds between the subsequent cohort of banks. The main role of this company is to keep track of the amortized-cost value of the loans to replicate the situation in which bonds were held by banks operating over multiple periods. At date t , this company buys the bonds at (real) market price Q_t^S from surviving banks that bought them at $t - 1$ and from the Deposit Insurance Agency (DIA), which repossesses the bonds from failing banks that bought them at $t - 1$. Then, the company sells the bonds at market price to the new cohort of banks that buy them (together with the newly issued long-term bonds) at t .

Importantly, this company provides a “certificate of amortized-cost value” to the bonds, which allows banks to write their balance sheet for regulatory purposes (in real terms) as follows:

$$L_{kt} + S_{kt} + Q_t^{AC} S_{kt}^L = \bar{B}_t + D_{kt} + (Q_t^{AC} - Q_t^S) S_{kt}^L, \quad (15)$$

where Q_t^{AC} is the real average amortized-cost value of bonds according to the certificate, and $(Q_t^{AC} - Q_t^S) S_{kt}^L$ measures what would be regarded as unrealized capital losses (if $Q_t^{AC} > Q_t^S$) or gains (if $Q_t^{AC} < Q_t^S$) if the bank were measuring the value of its bonds at their certified

in this manner, but cannot use it to avoid default due to its non-pecuniary nature.

amortized cost. Given the permanent inventory dynamics of the stock of long-term bonds, the law of motion of Q_t^{AC} is given by:

$$Q_t^{AC} S_{kt}^L = \frac{Q_{t-1}^{AC}}{\Pi_t} \left(1 - \frac{1}{m}\right) S_{kt-1}^L + Q_t^S \left[S_{kt}^L - \left(1 - \frac{1}{m}\right) S_{kt-1}^L \right], \quad (16)$$

where the first term in the right hand side represents the continuation amortized-cost value of the non-matured bonds, and the second term is the market value of the newly issued bonds acquired by the bond management company in the primary bond market.¹²

Capital requirement. Banks are subject to a minimum capital requirement, which imposes that banks must operate with *regulatory* equity capital greater than or equal to a fraction γ of their loans. The key novelty in our analysis is the comparison of capital requirements under two different definitions of regulatory capital. Under a fair-value definition, the capital requirement is of the following form:

$$\gamma L_{kt} \leq \bar{B}_t. \quad (17)$$

Under the amortized-cost definition, the requirement takes the following form:

$$\gamma L_{kt} \leq B_{kt}^{AC}, \quad (18)$$

where

$$B_{kt}^{AC} \equiv \bar{B}_t + (Q_t^{AC} - Q_t^S) S_{kt}^L \quad (19)$$

represents the amortized-cost value of bank equity, which is defined using Equation (15). Therefore, the difference between the two resulting capital requirements arises from the prudential treatment of *unrealized* gains and losses associated with banks' long-term bonds portfolio.

Terminal net worth of a bank. As derived in Equation (9), conditional on the island-idiosyncratic shock ω_k , an entrepreneurial firm pays back its loan in full when it experiences a firm-idiosyncratic shock no lower than $\bar{\omega}_{Ft+1}(\omega_k)$. In case of default of an entrepreneurial firm, the bank only recovers a fraction $(1 - \delta_M)$ of the firm's terminal asset value in Equation (8), where δ_M is an asset repossession cost. Hence, the nominal ex-post gross rate of return

¹²Note that the amortized-cost value of the newly issued bonds coincides with the market value.

on loans of the bank in island k is

$$\begin{aligned}\tilde{R}_{Lkt+1}(\omega_k) &= \frac{\omega_k(1 - \delta_M)\Pi_{t+1}[Q_{t+1}(1 - \delta) + z_{t+1}]K_t^E}{L_t} \int_0^{\bar{\omega}_{Ft+1}(\omega_k)} \omega_j dF_{jt+1}(\omega_j) \\ &\quad + R_{Lkt} \int_{\bar{\omega}_{Ft+1}(\omega_k)}^{\infty} dF_{jt+1}(\omega_j),\end{aligned}\tag{20}$$

where K_t^E denotes the aggregate level of physical capital held by entrepreneurs. By definition of the entrepreneurial firm's default threshold in Equation (9), the first term is bounded by R_{Lkt} : when borrowers default, they repay less than the agreed loan rate, otherwise they repay fully. This naturally limits upside-risk for the bank and therefore leads to a negatively skewed distribution of $\tilde{R}_{Lkt+1}(\omega_k)$.¹³

Due to the stochastic maturity of the long-term bond portfolio, the nominal gross rate of this portfolio is:

$$R_{kt+1}^S = \frac{\frac{1}{m} + (1 - \frac{1}{m}) \Pi_{t+1} Q_{t+1}^S}{Q_t^S}.\tag{21}$$

The nominal terminal net worth of the bank on island k is then

$$P_{t+1}\Omega_{kt+1}^B(\omega_k) = P_t \left[\tilde{R}_{Lkt+1}(\omega_k)L_{kt} + R_t S_{kt} + R_{kt+1}^S Q_t^S S_{kt}^L - c_f - R_{Dkt} D_{kt} \right].\tag{22}$$

Banks default on their deposits if their terminal net worth is negative. From Equation (22), it is useful to define a threshold value for the island-specific shock ω_k below which the bank in island k defaults. This is implicitly done in the next equation

$$\tilde{R}_{Lkt+1}(\bar{\omega}_{bkt+1})L_{kt} + R_t S_{kt} + R_{kt+1}^S Q_t^S S_{kt}^L - R_{Dkt} D_{kt} - c_f = 0.\tag{23}$$

Equation (23) implies that banks' failure rate at the beginning of period $t+1$ is $F_{kt+1}(\bar{\omega}_{kt+1})$. Thus, the nominal gross rate of return on the portfolio of equity of a banker that symmetrically invests in all banks is

$$\rho_{t+1}^B = \frac{\Pi_{t+1} \int_{\bar{\omega}_{bkt+1}}^{\infty} \Omega_{kt+1}^B(\omega_k) dF_{kt+1}(\omega_k)}{\bar{B}_t}.\tag{24}$$

¹³Thus the model features a structural link between bank asset returns and borrower defaults, which is crucial to accurately capture the importance of the feedback loop between conditions in the real economy and the financial sector (see Mendicino et al., forthcoming).

Bank's profit maximization problem. Banks are managed in the interest of bankers, and maximize:

$$\max_{L_{kt}, S_{kt}, S_{kt}^L, R_{Dkt}} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left\{ c_R L_{kt} + \int_0^\infty \max[\tilde{R}_{L_{kt+1}}(\omega) L_{kt} - R_{Dkt} D_{kt} + R_t S_{kt} + R_{t+1}^S Q_t^S S_{kt}^L - c_f, 0] F_{kt+1}(\omega) \right\} \quad (25)$$

subject to

$$L_{kt} + S_{kt} + Q_t^S S_{kt}^L = D_{kt} + \bar{B}_t, \quad (26)$$

$$D_{kt} = \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t, \quad (27)$$

and under fair-value capital requirements:

$$\gamma L_{kt} \leq \bar{B}_t, \quad (28)$$

while under amortized-cost requirements:

$$\gamma L_{kt} \leq B_t^{AC}. \quad (29)$$

The optimal deposit rate R_{Dt} is independent of all other choices, and is given by:

$$R_{Dkt} = \frac{\epsilon_D}{\epsilon_D - 1} R_t. \quad (30)$$

Hence, all banks offer the same deposit rate in every period. This implies $R_{Dt} = R_{Dkt}$, as well as $D_t = D_{kt}$. Since $\epsilon_D < -1$, the sensitivity of the deposit rate to the policy rate, $\frac{\epsilon_D}{\epsilon_D - 1}$ is below one, as in [Drechsler et al. \(2023\)](#). Hence, as in their model, the deposit rate is a fraction of the short-term rate R_t , such that the deposit spread is increasing with R_t .

Further, the problem implies the following arbitrage condition for the two bond types:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\frac{\frac{1}{m} + \left(1 - \frac{1}{m}\right) \Pi_{t+1} Q_{t+1}^S}{Q_t^S} - R_t \right] [1 - F_{kt+1}(\bar{\omega}_{bkt+1})]. \quad (31)$$

Furthermore, it can be proven that for any deposit rate (including the optimal deposit rate), the objective function is convex in the loan volume L_{kt} (but not necessarily strictly convex). Banks are indifferent between any $L_{kt} \in [0, \bar{L}_t]$ (where $\bar{L}_t = \frac{\bar{B}_t}{\gamma}$ under fair-value capital

requirements or $\bar{L}_t = \frac{B_t^{AC}}{\gamma}$ under amortized-cost capital requirements) if two conditions are simultaneously satisfied: if (i) a bank makes profits on non-lending activities that are so high that the bank never fails for any feasible loan volume $L < \bar{L}_t$, and (ii) the loan rate fulfills:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R + \int_{\bar{\omega}_{bkt+1}}^{\infty} [\tilde{R}_{L_{t+1}}(\omega) - R_t] dF_{kt+1}(\omega) \right] = 0. \quad (32)$$

Otherwise, there is a corner solution and either $L_{kt}^* = 0$ or the bank chooses the maximum loan volume it can extend without violating the capital requirement, i.e. $L_{kt}^* = \frac{\bar{B}_t}{\gamma}$ under fair-value based capital requirements, and $L_{kt}^* = \frac{B_t^{AC}}{\gamma}$ under amortized-cost based requirements. Which corner is optimal for the bank depends on the loan rate R_{Lkt} , which the bank takes as given. The proof can be found in the Appendix.

In summary, banks strictly prefer extending loans if the expected ex-post return is such that:

$$\begin{aligned} & \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \int_{\bar{\omega}_{bkt+1}}^{\infty} \left\{ [\tilde{R}_{L_{kt+1}}(\omega) - R_t(1 - \gamma)] \bar{L}_t + (R_t - R_{Dt}) D_t + (R_{t+1}^S - R_t) Q_t^S S_{kt}^L - c_f \right\} dF_{kt+1}(\omega) \\ & > \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} [(R_{t+1}^S - R_t) Q_t^S S_{kt}^L + (R_t - R_{Dt}) D_t - c_f + R_t \bar{B}_t - c_R \bar{L}_t]. \end{aligned} \quad (33)$$

This condition is verified numerically under the calibration explained in the next section. The role of the non-pecuniary benefit c_R is to make sure that under that calibration banks don't switch between extending zero loans and operating at maximum lending capacity, a pattern not observed in the data.¹⁴

Assuming this is the case, all banks choose identical loan volumes, as they receive the same level of equity from bankers. The k index is therefore dropped in continuation. In summary, bank payoffs at time t are a function of the maximum loan volume $\bar{L}_{t-1}$ the bank can extend at time $t - 1$ to comply with capital requirements, and are given by:

$$\begin{aligned} \Omega_t^B(\bar{L}_{t-1}) = & \int_{\bar{\omega}_{bt}}^{\infty} \left[(\tilde{R}_{L_{kt}}(\omega) - R_{t-1}(1 - \gamma)) \bar{L}_{t-1} - (R_{Dt-1} - R_{t-1}) D_{t-1} + (R_{t+1}^S - R_t) Q_t^S S_t^L - c_f \right] dF_{kt}(\omega). \end{aligned} \quad (34)$$

¹⁴If c_R were set to zero, equity would need to be made artificially scarce for (33) to hold in every period, implying counterfactually large loan spreads.

Deposit insurance agency. The DIA supervises the liquidation process of failed-bank assets, which is subject to proportional repossession costs δ_B .¹⁵ It imposes a nominal lump-sum tax T_{t+1} on bankers to (ex-post) balance its budget period-by-period. The total nominal lump sum tax T_{t+1} is

$$\begin{aligned} \frac{T_{t+1}}{P_t} &= [R_{Dt}D_t + c_f - R_{t+1}^S Q_t^S S_t^L - R_t S_t] F_{kt+1}(\bar{\omega}_{kt+1}) \\ &\quad - (1 - \delta_B) \left[\int_0^{\bar{\omega}_{bt+1}} \tilde{R}_{Lt+1}(\omega) L_t dF_{kt+1}(\omega) \right]. \end{aligned} \quad (35)$$

2.4 Contracting Problem Between Firms & Banks

Entrepreneurial firms enter a contract with the bank on their island k that specifies the loan rate and the leverage of entrepreneurs (or equivalently: the loan rate, the loan volume and the total amount of capital bought). Bankers are indifferent between any combination of loan rates and leverage on their iso-expected-profit curve:

$$\begin{aligned} \bar{\Omega}_t^b &= \\ \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \int_{\bar{\omega}_{bt+1}}^{\infty} &\left[(\tilde{R}_{Lt+1}(\omega) - R_t(1 - \gamma))L_t + (R_t - R_{Dt})D_t + (R_{t+1}^S - R_t)Q_t^S S_{kt}^L \right] dF_{kt+1}(\omega). \end{aligned} \quad (36)$$

Entrepreneurial firm jk active from time t to $t+1$ maximizes its properly discounted value for entrepreneurs $\mathbb{E}_t(\Lambda_{t,t+1}^E \Omega_{jkt+1}^{Firm})$, by choosing a point on the bank's isoprofit curve.¹⁶ From the perspective of the firms, the total loan volume L_t intermediated by each bank is exogenous (and given by either $\frac{B_t}{\gamma}$ or $\frac{B_t^{AC}}{\gamma}$ depending on the type of capital requirements). As in Mendicino et al. (forthcoming), firms also take the bank's default cutoff as given. Using Eq. (5), in equilibrium it must be that $s_t \bar{B}_t = \bar{\Omega}_t^b$. The contracting problem between the entrepreneurial firm and the bank on island k is then given by:

$$\begin{aligned} \max_{K_{jkt}, L_{jkt}, R_{Ljkt}} &\mathbb{E}_t \Lambda_{t,t+1}^E \Omega_{jkt+1}^{Firm} \\ \text{subject to Eq. (7), Eq. (36).} & \end{aligned} \quad (37)$$

¹⁵The model follows [Bernanke et al. \(1999\)](#) in adopting a “costly state verification” setup, by which the DIA must incur a cost that is proportional to the assets of the bank in order to observe the realization of the idiosyncratic shocks.

¹⁶There is no reason for entrepreneurial firms to make bankers better off than necessary for them to participate, therefore the contract they offer lies on the bank's iso-expected-profit curve.

The FOCs are presented in Appendix [A.7](#).

2.5 Fiscal Policy

As discussed above, there are two types of government bonds in this model: one-period bonds and long-term bonds. Both are assumed to be in fixed real supply: for the one-period bonds we assume a zero net-supply, while for the long-term bonds we assume a positive real supply S^L , a parameter to be calibrated. To introduce a demand side shock, we also assume the fiscal authority engages in government spending G_t , governed by the following AR(1) process:

$$\ln(G_t) = (1 - \rho_g)\ln(G) + \rho_g\ln(G_{t-1}) + \sigma_g\epsilon_{Gt}, \quad \epsilon_{Gt} \sim N(0, 1). \quad (38)$$

We assume that the government balances its budget in every period by charging households a lump-sum tax.

2.6 Monetary Policy

We assume that there is a central bank which sets the nominal gross policy rate R_t according to the following Taylor rule:

$$R_t = R^{1-\phi_R} R_{t-1}^{\phi_R} \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\phi_\pi(1-\phi_R)} \left(\frac{Y_t}{Y_{t-1}} \right)^{\phi_Y(1-\phi_R)} \tau_t, \quad (39)$$

where R is the long-term target monetary policy rate, and $\bar{\Pi}$ is steady state inflation. ϕ_R is a smoothing parameter, while ϕ_Y and ϕ_π govern how strongly the central bank reacts to deviations from GDP and inflation, respectively. τ_t is a monetary policy shock evolving according to

$$\ln(\tau_t) = \rho_\tau \ln(\tau_{t-1}) + \sigma_\tau \epsilon_{\tau t}, \quad \epsilon_{\tau t} \sim N(0, 1). \quad (40)$$

2.7 Production

The description of the production side of the economy follows a standard New Keynesian formulation and its full description is relegated to Appendix [A.3](#). Here, it shall suffice to state a few elements. The aggregate production function is

$$Y_t = \theta_t K_{t-1}^\alpha H_t^{1-\alpha}, \quad \text{with } \alpha \in [0, 1], \quad (41)$$

where K_t and H_t are aggregate physical capital and aggregate labor hours, respectively, and aggregate productivity θ_t is stochastic and follows an AR(1) process:

$$\ln(\theta_t) = \rho_\theta \ln(\theta_{t-1}) + \sigma_\theta \epsilon_{\theta t}, \quad \epsilon_{\theta t} \sim N(0, 1). \quad (42)$$

The New Keynesian Phillips Curve of the model, arising from the problem of a unit continuum of final good producers facing a stochastic elasticity of substitution between intermediate goods μ_t , as well as [Rotemberg \(1982\)](#) price adjustment costs is

$$\theta_R \left(\frac{\Pi_t}{\bar{\Pi}} - 1 \right) \frac{\Pi_t}{\bar{\Pi}} = \mathbb{E}_t \Lambda_{t,t+1} \theta_R \left(\frac{\Pi_{t+1}}{\bar{\Pi}} - 1 \right) \frac{\Pi_{t+1}}{\bar{\Pi}} \frac{Y_{t+1}}{Y_t} + mc_t \mu_t + (1 - \mu_t). \quad (43)$$

The stochastic elasticity of substitution μ_t also follows an AR(1) process:

$$\ln(\mu_t) = (1 - \rho_\mu) \ln(\mu) + \rho_\mu \ln(\mu_{t-1}) + \sigma_\mu \epsilon_{\mu t}, \quad \epsilon_{\mu t} \sim N(0, 1). \quad (44)$$

Physical capital is produced by combining the final good with undepreciated capital, subject to an adjustment cost of $\mathcal{C} \left(\frac{I_t}{K_{t-1}} \right)$ as in [Jermann \(1998\)](#).¹⁷ The aggregate capital stock evolves according to:

$$K_t = I_t + (1 - \delta) K_{t-1} + \mathcal{C} \left(\frac{I_t}{K_{t-1}} \right) K_{t-1}. \quad (45)$$

2.8 Capital Management Firms

There is a unit continuum of competitive capital management firms. They charge households a fee ς_t per unit of capital, and face costs of $\frac{\kappa_H}{2} (K_t^H)^2$. Their maximization problem is:

$$\max_{K_t^H} \varsigma_t K_t^H - \frac{\kappa_H}{2} (K_t^H)^2. \quad (46)$$

3 Solution, Estimation, and Model Validation

This section outlines the computational method used to obtain the numerical solution of the model, discusses the calibration strategy, and explores the quantitative properties of the

¹⁷The functional form is $\mathcal{C} \left(\frac{I_t}{K_{t-1}} \right) = \frac{a_{k,1}}{1 - \frac{1}{\kappa}} \left(\frac{I_t}{K_{t-1}} \right)^{1 - \frac{1}{\kappa}} + a_{k,2}$. While κ is estimated, $a_{k,1}$ and $a_{k,2}$ are set such that in steady state $I = \delta K$ and $\mathcal{C}(\delta) = 1$.

model.

3.1 Solution Method

The model is solved around the zero inflation steady state ($\bar{\Pi} = 1$), keeping in line with much of the New Keynesian literature (see [Galí, 2015](#)). We employ third-order perturbation methods to obtain an approximation of the policy functions around the deterministic steady state.¹⁸ The integrals involving the realized ex-post returns on bank loans (as well their derivatives) cannot be written as an explicit function of the state variables, which introduces a complication. We follow Mendicino et al. (forthcoming) in overcoming this challenge by approximating the integrals by a sum of third-order Taylor approximations. More details are provided in [Appendix A.7](#).

3.2 Model Estimation

The model is calibrated to quarterly euro-area data from 1995 Q1 to 2016 Q4. Following standard practices, the calibration of the model proceeds in two steps.¹⁹

First step. The coefficient of relative risk aversion σ is set to 1, which implies log-utility, the Frisch elasticity of labor supply φ_H to 1, the capital-share parameter of the intermediate goods production function α to 0.25, and the value of capital depreciation δ to 0.025. The elasticity of substitution parameters for differentiated labor services ϵ_W and final goods μ are set to 5 and 7.25, respectively, resulting in a wage markup of 20% ([Smets and Wouters, 2003](#)), and a markup of 16% in the goods market which is consistent with euro-area estimates reported in [Christopoulou and Vermeulen \(2012\)](#). The scaling parameter ξ_N associated with labor disutility is set to normalize a steady-state labor supply of $H = 1$. Following [Stähler and Thomas \(2012\)](#), the steady state government spending G is set to 22.56% of GDP. Following [Born and Pfeifer \(2014\)](#), we estimate the government spending shocks externally by OLS on Equation (38) (in logs).

¹⁸See [Christiano et al. \(2014\)](#), [Born and Pfeifer \(2014\)](#), and Mendicino et al. (forthcoming) for discussions on the necessity of third-order approximations to appropriately capture the effects of volatility shocks, such as those affecting the cumulative distribution functions of the firm-idiosyncratic and island-specific shocks.

¹⁹See e.g. [Fernández-Villaverde, Guerrón-Quintana, Rubio-Ramírez, and Uribe \(2011\)](#) and [Born and Pfeifer \(2014\)](#) for examples of DSGE models estimated in a two-step procedure using the Simulated Method of Moments.

The maturity of long-term bonds m is set to 13.6, which implies an average maturity of bank bond holdings of 3.4 years (Hoffmann, Langfield, Pierobon, and Vuillemeys, 2019). The value of bankruptcy parameters δ_B and δ_M are both set equal to 0.30, in line with empirical studies (e.g. Alderson and Betker, 1995; Djankov, Hart, McLiesh, and Shleifer, 2008; Granja, Matvos, and Seru, 2017). We set both θ_B and θ_E to 0.975, implying that bankers and entrepreneurs remain active for ten years on average. Finally, the minimum capital requirement γ is set to 0.08, consistent with the requirement under Basel II.

Second step. The remaining parameters are estimated using the Simulated Method of Moments (SMM).^{20,21} We obtain the model implied moments by simulating our baseline economy, under which banks' regulatory capital is defined on the basis of the amortized-cost of their bonds portfolio. Our estimation targets include a number of macroeconomic, financial, and banking moments. We target the standard deviations and first two autocorrelations of GDP, consumption, investment, inflation, wages, the policy rate and labor hours, as well as their correlation with GDP. We also target a range of moments related to financial markets. These are the mean and standard deviation of the conditional expectation of firm and bank default rates and the unconditional correlation between the two default probabilities, the mean and standard deviation of the loan rate spread, the average deposit rate spread, the average central bank policy rate, the average aggregate loan to GDP ratio, the share of physical capital owned by households, and the average ratio of loans to bonds on bank's balance sheets. Finally, we target the size of GDP contractions after a large decrease in bank equity: Baron, Verner, and Xiong (2021) report an average 4% equity decline within a year after a 30% drop in bank equity.

Tables 1 and 2 provide the values of moments targeted in the data, and compare them to their model generated counterparts. Parameters value are reported in the Appendix (Table 5). We obtain data on GDP, consumption, investment, government spending, total wages, hours worked, the GDP deflator and population from the OECD Quarterly National Accounts. Data on financial corporation loan volumes, loan rates and riskless interest rates are obtained from the ECB Statistical Data Warehouse.²² All series are adjusted for seasonality

²⁰The set of estimated parameters is $\beta, \xi_B, \xi_E, \sigma_{\omega_k}, \rho_{\omega_k}, \rho_{\omega_j}, \sigma_{\omega_j}, \bar{\sigma}_{\omega_k}, \bar{\sigma}_{\omega_j}, \sigma_\theta, \rho_\theta, \sigma_\tau, \rho_\tau, \sigma_\mu, \rho_\mu, b, \theta_R, \theta_W, \kappa, \kappa_H, \phi_R, \phi_\Pi, \phi_Y, S^L$.

²¹The good properties of SMM for estimation of non-linear DSGE models have been established in Ruge-Murcia (2012).

²²The loan volume series include all available maturities. The loan spread is constructed as a volume weighted average over maturities, where the riskless rate is taken as the short term rate published by the ECB for maturities less than 1 year, the 2-year yield on triple A euro-area government bonds (published by the ECB) for maturities between 1 and 5 years, and the corresponding 5-year yield for maturities over 5

Table 1: *Calibration targets and model fit (macroeconomic)*

	$\sigma_{x_t}/\sigma_{\Delta y_t}$ (Y: $\sigma_{\Delta y_t}$)		$\rho(x_t, \Delta y_t)$	
	Data	Model	Data	Model
ΔY	1.186	1.452	1	1
ΔC	0.746	0.77	0.915	0.672
ΔI	2.594	2.835	0.93	0.693
Π	0.229	0.493	0.326	0.296
Δw	0.478	0.548	-0.164	-0.318
R	0.308	0.399	0.489	0.449
H	1.828	3.332	0.217	0.605
	$\rho(x_t, x_{t-1})$		$\rho(x_t, x_{t-2})$	
	Data	Model	Data	Model
ΔY	0.905	0.829	0.705	0.637
ΔC	0.888	0.928	0.702	0.778
ΔI	0.872	0.956	0.754	0.851
Π	0.437	0.69	0.372	0.515
Δw	0.868	0.846	0.787	0.607
R	0.971	0.831	0.922	0.66
H	0.915	0.969	0.837	0.927

Notes: All series are seasonally adjusted and all variables except Π and R are in logs. Δ^{HP} indicates the Hodrick-Prescott filter with HP parameter 1600. Data sources and variable definitions are described in Appendix B.

and those series that exhibit trends are detrended using the Hodrick-Prescott filter.²³ The moments for the mean and standard deviation of firm and bank defaults, the correlation between firm and bank defaults, and the mean and standard deviation of the rate of return on bank equity are taken from Mendicino et al. (forthcoming).²⁴ Appendix B contains further details on the data sources and construction, and the calibration strategy.

The model fits the data well. It struggles to match the relative volatility of hours worked, which is not surprising given that the model contains no labor market frictions.

years.

²³Given that our data is quarterly, following Ravn and Uhlig (2002), we set the HP parameter to 1600. For those series that are not directly available with seasonal adjustment, such as loans, seasonal adjustment is done using X13-ARIMA using the R package seasonal.

²⁴The moments are for the euro area and almost the same time period: Mendicino et al. (forthcoming) use data from 1992Q1:2016Q4

Table 2: Calibration targets and model fit (financial)

Moment	Description	Data	Model
$100\mathbb{E}_t \left(\int_0^\infty F_{jt+1}(\bar{\omega}_{jt+1}(\omega_k))d\omega_k \right)$	Mean Firm Default	0.662	0.746
$100\mathbb{E}F_{kt+1}(\bar{\omega}_{kt+1})$	Mean Bank Default	0.166	0.209
$\rho \left(\int_0^\infty F_{jt+1}(\bar{\omega}_{jt+1}(\omega_k))d\omega_k, F_{kt+1}(\bar{\omega}_k) \right)$	Corr(Firm D., Bank D.)	0.642	0.602
$100\sigma \left(\int_0^\infty F_{jt+1}(\bar{\omega}_{jt+1}(\omega_k))d\omega_k \right)$	Std Firm Def.	0.549	0.362
$100\sigma (F_{kt+1}(\bar{\omega}_{kt+1}))$	Std Bank Def.	0.422	0.467
$100\sigma_{\rho_t^B}$	Std ROE Banks	2.06	3.26
$100(\mathbb{E}\rho_t^B - 1)$	Mean ROE Banks	1.6	2.64
$\mathbb{E}\frac{L_t}{Y_t}$	Mean L/Y	2.35	2.41
$100\mathbb{E}(R_{Lt} - R_t)$	Mean Loan spread	0.549	0.167
$\sigma_{(R_{Lt}-R_t)}/\sigma_{\Delta Y_t}$	Sd Loan spread	0.137	0.101
$\mathbb{E}R_t$	Mean Policy Rate	1.004	1.004
$\mathbb{E}(L_t/S_t^L)$	Loan-to-Bond Ratio	3.6	3.62
$\mathbb{E}(K_{Ht}/K_t)$	HH Capital Share	0.22	0.232
$400\mathbb{E}(R_t - R_{Dt})$	Deposit Spread	1	1
$\mathbb{E}\frac{\Delta_4 \log(Y_{t+4})}{\Delta_4 \log(EQ_t)} \mid p_1(\Delta \log(EQ_t))$	GDP Loss From Large Equity Loss	-0.133	-0.148

Notes: All series are seasonally adjusted. Δ^{HP} indicates the Hodrick-Prescott filter with HP parameter 1600. $\Delta_4 x_t$ indicates the fourth difference (that is, one-year difference) $x_t - x_{t-4}$, and $p_1(x)$ indicates that x is below it's first percentile. Data sources and variable definitions are described in Appendix B.

3.3 Model Validation

In this section, we validate the performance of our model by assessing it against the empirical literature on loan pricing implications of realized and unrealized bank losses.

In a recent study, Volk (2024) finds that “banks with 1 pp higher share of unrealized losses in their risk-weighted assets charge on average 8 bps higher corporate lending rate in Slovenia. These unrealized losses have a lower impact compared to actual changes in capital, for which the literature establishes the impact of around 10-25 bps.” We simulate our baseline model economy, i.e. the model in which banks are subject to amortized cost requirements, for 100,000 periods and define banks’ unrealized losses as the percentage difference between amortized-cost value equity and fair value equity:

$$Unrealized_t = 100 \frac{B_t^{AC} - B_t}{mean(B^{AC})}. \quad (47)$$

The actually realized losses in capital are defined as the shortfall of bank equity below its

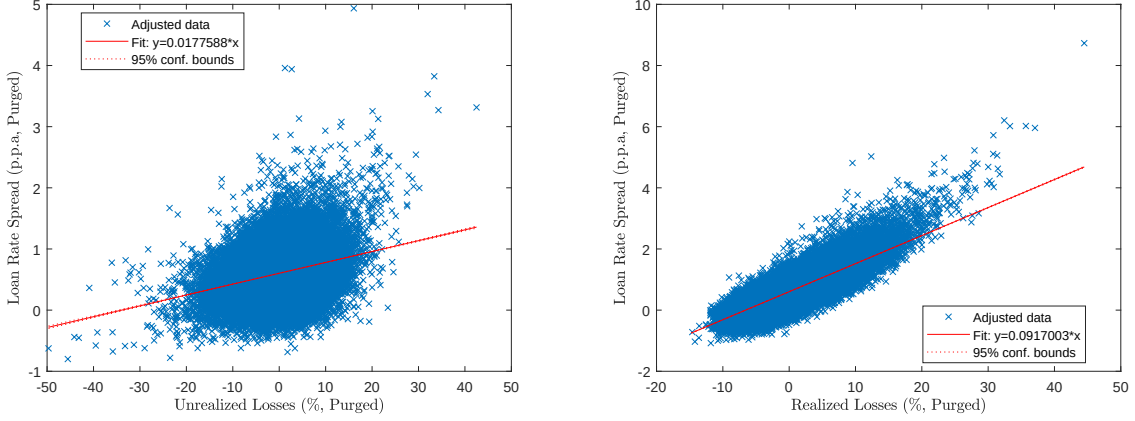


Figure 1: *Loan pricing effects: realized vs. unrealized losses*

Notes: The figure presents coefficients from the linear regression in Equation (49). Both equations are estimated on simulated data from the baseline model.

average:

$$Realized_t = 100 \frac{mean(B^{AC}) - B_t^{AC}}{mean(B^{AC})} \quad (48)$$

We then investigate the loan pricing implications of realized and unrealized bank losses using linear regressions of the following form:

$$spread_t = \beta_0 + \beta_1 Unrealized_t + \beta_2 Realized_t + \beta_3 K_t + \beta_4 Q_t + \beta_5 E_t, \quad (49)$$

where the dependent variable is the *spread* between the loan rate and the policy rate (defined in annualized percentage points, i.e., $spread_t = 400(R_{Lt} - R_t)$), and where we control for aggregate physical capital and its price, as well as firm equity.

The results are depicted in Figure 1. In line with the literature (although slightly lower) we find that unrealized losses on banks' balance sheet have a significant positive impact on loan pricing ($\beta_1 \approx 0.018$, corresponding to an approximately 1.8 bps increase in the annualized loan rate), and that this response is weaker than the one associated with actual changes in bank capital ($\beta_2 \approx 0.092$, corresponding to an approximately 9.2 bps increase in the annualized loan rate).

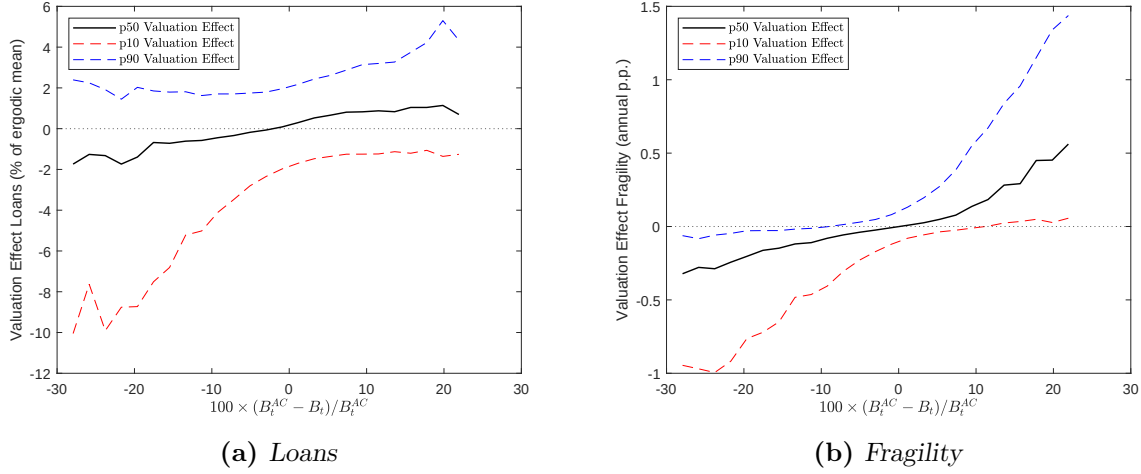


Figure 2: Valuation Effects

Notes: The left panel of this figure shows the Valuation Effect on Loans, defined as the difference in bank loans between the baseline economy (amortized-cost regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The difference in bank loans is reported as a percentage of the ergodic mean of loans under the amortized-cost regime. The right panel of this figure shows the Valuation Effect on Fragility, defined as the difference in the annualized bank default probability between the baseline economy (amortized-cost regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The p50, p10 and p90 Valuation Effects are defined as, respectively, the 50th, 10th and 90th percentile of the relevant differences conditional on the level of unrealized losses or gains.

4 The Effects of Capital Requirements

In this section, we analyze the performance of the economy under alternative regulatory accounting frameworks, identifying the approach that maximizes social welfare. Throughout the analysis, we compare endogenous responses in the baseline economy, with an economy in which banks' regulatory capital is defined on the basis of the fair value of the bonds, using an identical sequence of exogenous shocks.

4.1 Credit Supply and Fragility

We first assess the effects that prudential treatment of unrealized gains and losses has on credit supply and bank default probabilities. To do so, we perform the following exercise. Simulating the baseline economy for 100,000 periods, we compute loan quantity and bank failure probability for different levels of unrealized losses or gains on banks' balance sheet, defined as $(B_t^{AC} - B_t) / B_t^{AC}$. Then, we simulate an alternate economy in which regulatory

capital is defined on the basis of the fair value of bonds, where we compute the same variables for different *hypothetical* levels of unrealized losses or gains, i.e., we compute the losses or gains that a bank would have accumulated on its balance sheet, were it subject to the amortized-cost approach.

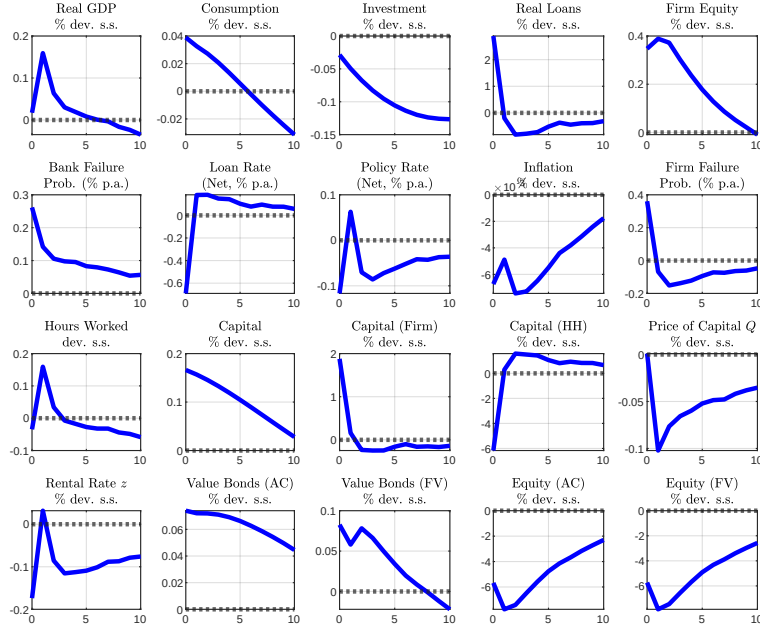
The left panel of Figure 2 depicts the Valuation Effect on Loans as the difference in bank loans between the baseline economy (amortized-cost regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The difference in bank loans is reported as a percentage of the ergodic mean of loans under the amortized-cost regime. The right panel of Figure 2 depicts the Valuation Effect on Fragility, which we define as the difference in the annualized bank default probability between the baseline economy (amortized-cost regime) and the alternate economy (fair-value regime), for different levels of unrealized losses or gains. These figures show that when unrealized losses in banks' portfolio accumulate over time, their default probabilities are higher in the amortized-cost regime. At the same time, banks extend more loans in this situation as they are relatively less constrained by the capital requirement prevailing in the amortized-cost regime – banks operate with a higher effective leverage. Considering that the average annual bank failure probability is 0.66%, the results are economically significant, with a median valuation effect on fragility of about 0.2 p.p. for accumulated unrealized losses amounting to 10% of regulatory equity.

4.2 Anatomy: Episodes with Large Accumulated Unrealized Losses

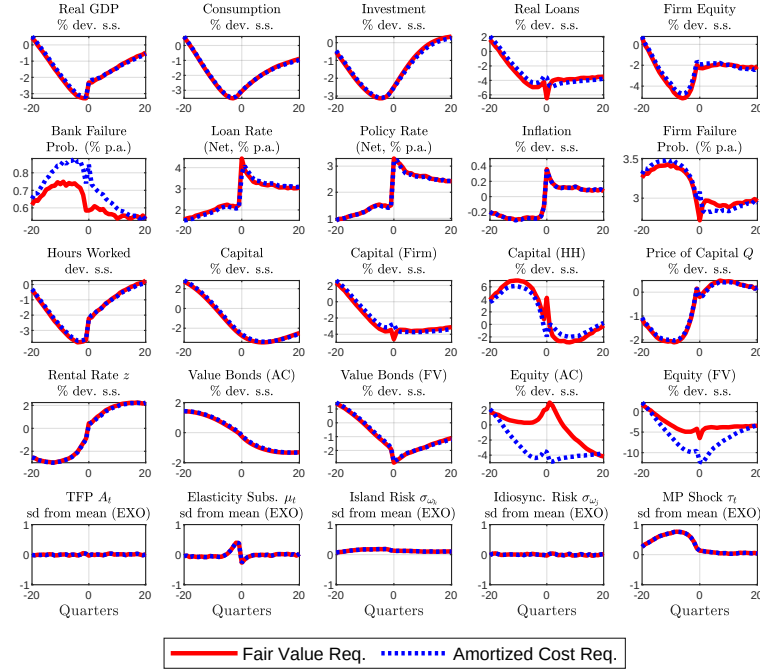
To understand the results, we plot the simulated time series around episodes in which banks' balance sheet contain large accumulated unrealized losses. The criteria for these episodes is that $AL_t \equiv B_t^{AC} - B_t$ exceeds its 90th percentile, denoted by $p_{90}(AL)$. Let $t = 0$ be the first period of such an episode: $AL_{-1} < p_{90}(AL)$ and $AL_0 \geq p_{90}(AL)$. It is crucial to note that we use an identical sequence of exogenous shocks across the simulations in both regulatory regimes (as well as identical to those used in all previous exercises). Nevertheless, macroeconomic conditions *endogenously* differ in the two regimes — *exclusively* due to the difference in regulatory treatment of unrealized gains and losses on banks' bonds portfolio.

Figure 3 shows the average path leading to large accumulated unrealized losses. The figure depicts both endogenous and exogenous variables (labeled EXO).

The average event with large accumulated unrealized losses is preceded by a hump-shaped evolution of the monetary policy shock, peaking several quarters before the event and gradually fading back toward zero by the time the average episode begins. This monetary tighten-



(a) Differences (Amortized Cost - Fair Value)



(b) Levels

Figure 3: Anatomy: Episodes with large accumulated unrealized losses

Notes: Time 0 denotes the start of a spell in which banks' balance sheet contain large accumulated unrealized losses, such that $AL_{-1} < p_{90}(AL)$ and $AL_0 \geq p_{90}(AL)$, where $AL_t \equiv B_t^{AC} - B_t$. Exogenous variables are labeled as EXO.

ing phase plunges the economy into a recession: GDP, inflation, consumption, and investment decline persistently. In the financial sector, tightening monetary policy generates unrealized losses from long-term bonds on the representative bank's balance sheet. Furthermore, shortly before the event, a cost-push shock (a temporary increase in the elasticity of substitution across intermediate goods producers) builds up and peaks prior to the event, before falling sharply at the event date. This causes inflation to rise sharply at the event itself, triggering a strong endogenous policy rate hike through the Taylor rule. This large increase in the policy rate further reduces the market value of long-term government bonds held by banks, causing the gap between the book and market value of equity to widen sharply.

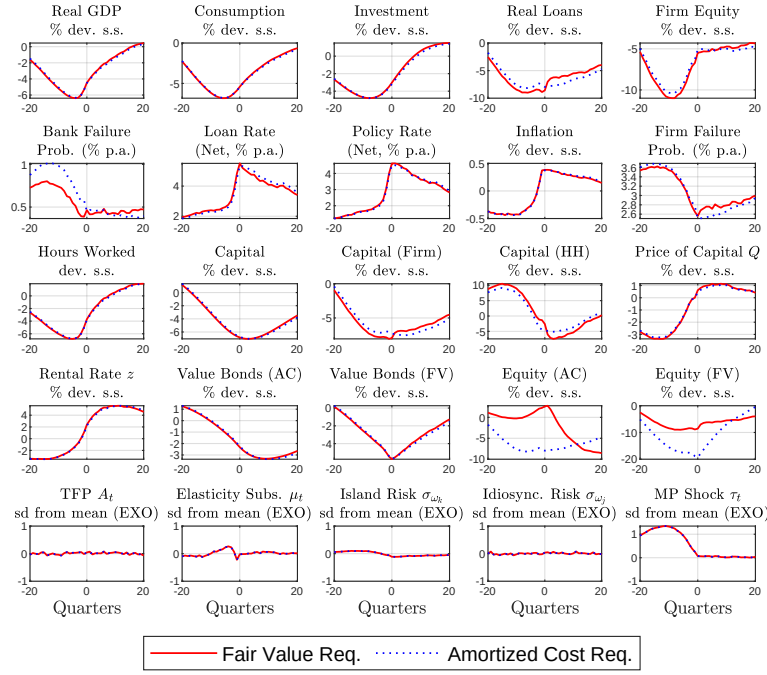
The recession peaks shortly before the event itself. During the final pre-event periods, consumption and investment begin to recover as the direct effects of the earlier monetary tightening dissipate. While GDP and hours worked increase at the event date, consumption and investment continue their smooth recovery path without a visible discontinuity. The temporary increase in GDP is driven by higher price adjustment costs associated with the inflation spike, rather than by an expansion in private demand ($C_t + I_t$).

In summary, these episodes reflect the interaction of previously accumulated valuation losses on bank balance sheets with a sharp, cost-push inflation driven, monetary policy tightening that aggravates these valuation losses. These dynamics are broadly consistent with the post-COVID tightening cycle, during which supply-side inflationary pressures and aggressive monetary policy hikes led to substantial valuation losses on banks' bond portfolios (Marsh and Laliberte, 2023).²⁵

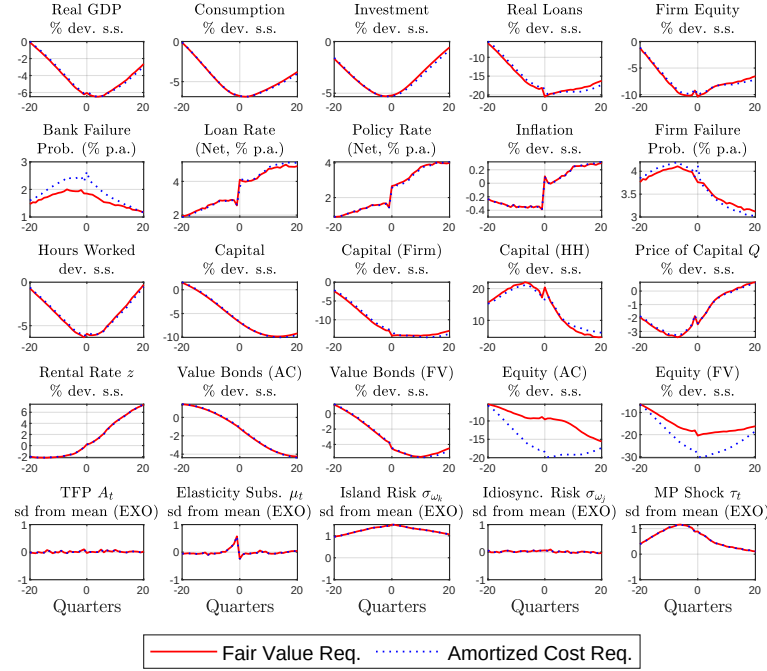
The evolution of loan supply and financial stability during these episodes then depend on the regulatory accounting framework. In the amortized-cost regime, banks are able to maintain a higher loan supply compared to the fair-value regime. However, since the solvency of banks depends on the fair value of their assets, this means that banks operate with effectively higher leverage, which is associated with a gradual increase in their default probability during the policy rate tightening vis-a-vis the fair-value accounting regime.

During the subsequent transition back to the steady state, loan supply is actually higher under fair-value accounting. This is because the amortized-cost value of equity is far more sluggish than the fair value of equity.

²⁵The historical episode also involved strong demand components and fiscal stimulus not captured in the model (in which government spending is exogenous).



(a) Valuation effect on fragility in Q1



(b) Valuation effect on fragility in Q4

Figure 4: Anatomy: The role of credit risk

Notes: Time 0 denotes the start of a spell in which banks' balance sheet contain large accumulated unrealized losses, such that the fair value of equity is between 10 and 30% below the amortized cost value. Exogenous variables are labeled as EXO.

4.3 The Role of Credit Risk

The valuation effects plotted in Figure 2 suggest that there are several paths the economy can take over the monetary policy cycles. For instance, the median valuation effect on fragility for 20% accumulated unrealized losses is below 0.5 percentage points, while the 90th percentile effect for the same level of accumulated losses is close to 1.5 percentage points.

To understand these differences, we plot the simulated time series around episodes in which banks' balance sheet contain large accumulated unrealized losses, conditioning on the distribution of the valuation effects. The left (right) panel of Figure 4 shows the evolution of the economy in a window during which the valuation effect on fragility is in its first (fourth) quartile. These figures show that the underlying non-linearities in the valuation effects of fragility (for a given level of unrealized losses) are largely driven by the realization of island-risk shocks. When island-risk is high (i.e., the island-specific shock, which banks cannot diversify away, has a high variance) banks have a significantly higher default probability than when island risk is low, independent of the accounting regime. In the context of this increased bank riskiness, the increase in the bank default probability due to the effectively higher leverage allowed by capital requirements based on the amortized-cost value of equity in the presence of unrealized losses in the representative bank's bond portfolio is particularly large.

4.4 Welfare-Maximizing Regulatory Accounting Approach

We now turn to identifying the regulatory accounting approach that maximizes social welfare in our model. We assess both whether regulatory capital should be defined on the basis of the amortized-cost or fair value of equity, and how high the capital requirements should be. We find that regulatory capital defined on the basis of the fair-value of the bonds is overall superior in terms of welfare to defining it on the basis of their amortized-cost.

Table 3 reports means and standard deviations of important financial and real economy variables by accounting regime. Additionally, Figure 5 shows how selected variables change with the level of the capital requirement in each of the regimes.

On average, banks supply slightly lower loan volumes at slightly lower spreads under fair value requirements. The default probability of both banks and firms is also slightly lower under fair-value requirements. In the case of firms, this is since lower aggregate lending translates into lower leverage. This effect is reinforced by the feedback loop between lower

Table 3: *Summary Statistics by Accounting Regime*

Variable	Mean		SD	
	FV	AC	FV	AC
Loans (L)	4.117	4.1188	0.5135	0.5128
Bank Defaults p.a. ($400 \times F_{kt}(\bar{\omega}_{bt})$)	0.5198	0.5339	0.934	0.9859
Firm Defaults p.a ($400 \times \int_0^\infty F_{jt}(\bar{\omega}_{jt}(\omega_k))dF_{kt}$)	3.001	3.0071	1.0448	1.0563
Consumption (C)	1.0789	1.0787	0.0534	0.0535
Total Production (Y)	1.7277	1.7281	0.0859	0.0864
Inflation (Π)	0.9971	0.9971	0.0061	0.0062
Wage Inflation (Π_W)	0.9971	0.9971	0.0044	0.0044

loan rates and lower firm default probabilities. Lower firm default probabilities then contribute to lower bank default probabilities. In combination, this implies that a lower part of aggregate output has to be used to cover the deadweight loss of defaults and in consequence a higher part is available for consumption. This explains why average consumption is higher under fair value requirements, despite the slightly lower average aggregate physical capital under that accounting regime. On average higher (and slightly less volatile) consumption translates into welfare gains from fair-value accounting.²⁶ Our results thus lend support to the Basel III proposal of recognizing unrealized gains and losses in regulatory capital (i.e., removing prudential filters).

The differences in utility between both accounting regimes are moderate in magnitude. For an 8% capital charge, the gains from amortized-cost measure of regulatory capital are approximately 2.7 bps. Regarding the size of capital requirements, we find an optimal value of 13% – well above the Basel II capital charge, but below the 16% found in Mendicino et al. (forthcoming). Optimal (fair-value) capital requirements of 13% would be associated with an approximately 15 bps increase in consumption equivalent utility terms.

²⁶In a New Keynesian setup without deadweight losses from default, [Benigno and Woodford \(2003\)](#) show that welfare is a function of the mean and variance of output (Y), price and wage inflation (Π , Π_W). In our model, there is an additional wedge between consumption and output stemming from default deadweight losses. Differences in the mean and standard deviation of price and wage inflation inflation are negligible in our results.

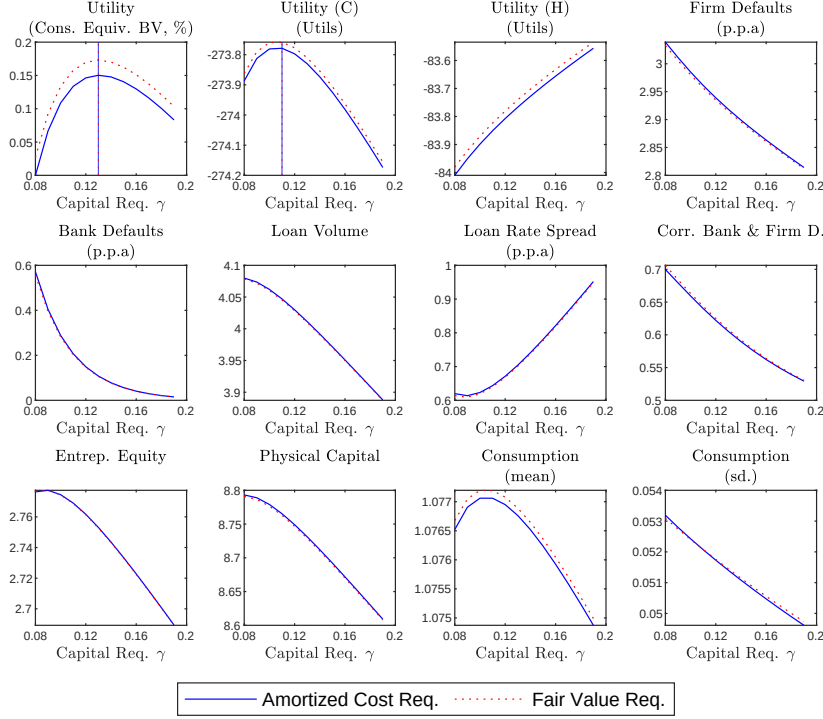


Figure 5: *Optimal capital requirements*

Notes: Utility is household utility, which is the relevant welfare benchmark as all banks and firms are owned by household. Utility (C) and Utility (H) are the parts attributed to consumption and labor, respectively. The benchmark for computing consumption equivalents is the baseline model with a capital requirement of 8%. All parameters other than the capital requirement γ are kept fixed.

5 Conclusion

This paper examines how the regulatory capital treatment of unrealized capital gains and losses on banks' debt securities, related to the impact of interest rate risk, affects financial stability and credit supply. To this purpose, it develops a dynamic general equilibrium model in which banks are exposed to both interest rate risk and credit risk.

We find that having regulatory capital defined on the basis of the fair-value of the bonds (as proposed in Basel III) is overall slightly superior in terms of welfare to having regulatory capital defined on the basis of their amortized-cost value. Measuring at fair value makes regulatory capital more sensitive to changes in the interest rate. This translates to a greater volatility in credit. At the same time, the terms of the financial contract between banks and firms are better aligned with macro-economic conditions and bank balance sheet fundamentals. The better pricing of risk makes both banks and firms safer on average, freeing up

resources otherwise spent on deadweight losses from bankruptcies. This results in an overall higher level and lower volatility of consumption under fair-value accounting.

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Appendices

A Model Details

This Appendix describes elements of the standard New Keynesian models omitted in Section 2 of the main text, and presents the details of the equilibrium.

A.1 Household Problem

Households maximize Equation (1) subject to the budget constraint Equation (2). We begin by providing details of the components of household's cash flows, summarized by Σ_t in Equation (2). Households:

- receive net payoffs $(1 - \theta_B)(1 - \xi_B)\rho_t^B P_{t-1} B_{t-1}$ from bankers,
- receive net payoffs $(1 - \theta_E)(1 - \xi_E)\rho_t^E P_{t-1} E_{t-1}$ from entrepreneurs,
- receive $(P_t - P_t^m)Y_t - \frac{\theta_R}{2}(\Pi_t - 1)^2 P_t Y_t$ from final good producers (see Section A.3),
- receive profits $P_t \Pi_t^C$ from capital producers and $P_t(\varsigma_t K_t^H - \frac{\kappa_H}{2}(K_t^H)^2)$ from capital managers (see Section A.4),
- are charged a lump sum fee $\frac{\theta_W}{2}(\Pi_t^W - 1)^2 W_t$ from a labor union that negotiates wages (explained in the next subsection),
- are charged lump sum taxes LT_t by the government.

Therefore:

$$\begin{aligned} \Sigma_t = & (1 - \theta_B)(1 - \xi_B)\rho_t^B P_{t-1} B_{t-1} + (1 - \theta_E)(1 - \xi_E)\rho_t^E P_{t-1} E_{t-1} + P_t(\varsigma_t K_t^H - \frac{\kappa_H}{2}(K_t^H)^2) \\ & + P_t \Pi_t^C + (P_t - P_t^m)Y_t - \frac{\theta_R}{2}(\Pi_t - 1)^2 P_t Y_t - \frac{\theta_W}{2}(\Pi_t^W - 1)^2 W_t - LT_t. \end{aligned} \quad (\text{A.1})$$

The multiplier on the budget constraint Equation (2) (which is expressed in nominal terms) is $\tilde{\lambda}_t$. We can then define the multiplier on the real budget constraint as $\lambda_t = P_t \tilde{\lambda}_t$ and express the real stochastic discount factor as

$$\Lambda_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t}. \quad (\text{A.2})$$

We obtain the first order conditions for consumption, capital, deposits, and riskless one-period bonds, as follows:

$$(C_t - bC_{t-1})^{-\sigma} - \beta b \mathbb{E}_t(C_{t+1} - bC_t)^{-\sigma} = \lambda_t, \quad (\text{A.3})$$

$$\mathbb{E}_t \Lambda_{t,t+1} [z_{t+1} + (1 - \delta)Q_{t+1}] = Q_t + \varsigma_t, \quad (\text{A.4})$$

$$\mathbb{E}_t \Lambda_{t,t+1} \frac{R_{Dt}}{\Pi_{t+1}} = 1, \quad (\text{A.5})$$

$$\mathbb{E}_t \Lambda_{t,t+1} \frac{R_t}{\Pi_{t+1}} = 1. \quad (\text{A.6})$$

As explained in the main text, the labor supply decision is relegated to a labor union, whose problem is described in continuation.

A.2 Wage Setting

Wage setting is subject to [Rotemberg \(1982\)](#) adjustment costs governed by parameter θ_W which the labor union finances by charging households a lump-sum fee. No costs arise from adjusting wages according to the steady-state inflation $\bar{\Pi}$.²⁷ The labor packer's demand for variety h is:

$$H_{ht} = \left(\frac{W_{ht}}{W_t} \right)^{-\epsilon_W} H_t. \quad (\text{A.7})$$

The labor union maximizes household utility subject to labor demand (Equation A.7 with Lagrange multiplier mrs_t) and the nominal household budget constraint (Eq. 2 with multiplier λ_t/P_t):

$$\begin{aligned} \max_{H_{ht}, W_{ht}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t & \left[\lambda_t \left(\frac{W_{ht}}{P_t} H_{ht} - \frac{\theta_W}{2} \left(\frac{W_{ht}}{\bar{\Pi} W_{ht-1}} - 1 \right)^2 \frac{W_t}{P_t} \right) \right. \\ & \left. - \frac{\xi_H H_{ht}^{1+\varphi_H}}{1+\varphi_H} - mrs_t \left(H_{ht} - \left(\frac{W_{ht}}{W_t} \right)^{-\epsilon_W} H_t \right) \right]. \end{aligned} \quad (\text{A.8})$$

We obtain the first order conditions for hours worked and wages as follows:

$$mrs_t = \lambda_t \frac{W_{ht}}{P_t} - \xi_H H_{ht}^{\varphi_H}, \quad (\text{A.9})$$

²⁷This may be thought of as wages being fully indexed to steady-state inflation: If the union does not actively adjust nominal wages, they grow by the steady-state inflation rate.

$$\begin{aligned} \theta_W \lambda_t \left(\frac{W_{ht}}{\bar{\Pi} W_{ht-1}} - 1 \right) \frac{W_t}{P_t} \frac{1}{\bar{\Pi} W_{ht-1}} &= \lambda_t \frac{H_{ht}}{P_t} - \epsilon_W m r s_t \left(\frac{W_{ht}}{W_t} \right)^{-\epsilon_W - 1} \frac{H_t}{W_t} \\ &- \beta \mathbb{E}_t \left[\lambda_{t+1} \theta_W \left(\frac{W_{ht+1}}{\bar{\Pi} W_{ht}} - 1 \right) \frac{W_{t+1}}{P_{t+1}} \frac{W_{ht+1}}{\bar{\Pi} W_{ht}^2} \right]. \end{aligned} \quad (\text{A.10})$$

Since the problem is identical for each variety h , there is no price dispersion and hence $W_{ht} = W_t$. Define the nominal wage inflation as $\Pi_t^W = \frac{W_t}{W_{t-1}}$. Then, the first order conditions can be combined to obtain the New Keynesian Wage Phillips Curve:

$$\theta_W \left(\frac{\Pi_t^W}{\bar{\Pi}} - 1 \right) \frac{\Pi_t^W}{\bar{\Pi}} = \theta_W \mathbb{E}_t \left[\Lambda_{t,t+1} \left(\frac{\Pi_{t+1}^W}{\bar{\Pi}} - 1 \right) \frac{(\Pi_{t+1}^W)^2}{\Pi_{t+1} \bar{\Pi}} \right] + (1 - \epsilon_W) H_t + \epsilon_W \frac{\xi_H H_t^{1+\varphi_H}}{\lambda_t w_t}, \quad (\text{A.11})$$

where

$$w_t = \frac{w_{t-1} \Pi_t^W}{\Pi_t} \quad (\text{A.12})$$

is the law of motion of real wages.

A.3 Production

A representative competitive intermediate good producer has access to Cobb-Douglas production technology:

$$Y_t^m = \theta_t K_{t-1}^\alpha H_t^{1-\alpha}, \quad (\text{A.13})$$

where $\alpha \in [0, 1]$, and θ_t is an aggregate productivity shock that evolves according to an AR(1) process:

$$\log(\theta_t) = \rho_\theta \log(\theta_{t-1}) + \sigma_\theta \epsilon_{\theta t}, \quad (\text{A.14})$$

with $\epsilon_{\theta t} \sim N(0, 1)$. Denote the price of the intermediate good by P_t^m . The profit maximization problem of the representative intermediate good producer is then:

$$\max_{H_t, K_{t-1}} P_t^m \theta_t K_{t-1}^\alpha H_t^{1-\alpha} - P_t z_t K_{t-1} - P_t w_t H_t. \quad (\text{A.15})$$

Further, define the real price of the intermediate good. as $\frac{P_t^m}{P_t} = mc_t$. Then, the profit maximization problem yields the following FOCs:

$$mc_t \alpha \frac{Y_t^m}{K_{t-1}} = z_t \quad (\text{A.16})$$

$$mc_t (1 - \alpha) \frac{Y_t^m}{H_t} = w_t \quad (\text{A.17})$$

To incorporate nominal price rigidities, we model a unit continuum of monopolistic final good producers, each producing a differentiated variety i using a linear technology with the intermediate good as the only input:

$$Y_{it} = Y_t^m(i). \quad (\text{A.18})$$

The final good composite is the CES aggregate:

$$Y_t = \left(\int_0^1 Y_{it}^{\frac{\mu_t-1}{\mu_t}} di \right)^{\frac{\mu_t}{\mu_t-1}}, \quad (\text{A.19})$$

where μ_t is a stochastic elasticity of substitution that follows an AR(1) process:

$$\ln(\mu_t) = (1 - \rho_\mu) \ln(\mu) + \rho_\mu \ln(\mu_{t-1}) + \sigma_\mu \epsilon_{\mu t}, \quad (\text{A.20})$$

where $\epsilon_{\mu t} \sim N(0, 1)$ is an exogenous markup shock. Final good producers are subject to Rotemberg adjustment costs, governed by parameter θ_R , from steady state inflation $\bar{\Pi}$. They discount real payoffs by the household's real discount factor $\beta^t \lambda_t$. Their maximization problem in real terms is then:

$$\max_{P_t(i)} \mathbb{E}_t \sum_{n=0}^{\infty} \beta^n \frac{\lambda_{t+n}}{\lambda_t} \left[\frac{P_{t+n}(i)}{P_{t+n}} Y_{it+n} - mc_{t+n} Y_{it+n} - \frac{\theta_R}{2} \left(\frac{P_{t+n}(i)}{\bar{\Pi} P_{t+n-1}(i)} - 1 \right)^2 Y_{t+n} \right] \quad (\text{A.21})$$

$$\text{s.t. } Y_{it} = \left(\frac{P_t(i)}{P_t} \right)^{-\mu_t} Y_t \quad (\text{A.22})$$

The first order condition is:

$$\begin{aligned} (1 - \mu_t) \left(\frac{P_t(i)}{P_t} \right)^{-\mu_t} \frac{Y_t}{P_t} + mc_t \mu_t \left(\frac{P_t(i)}{P_t} \right)^{-\mu_t-1} \frac{Y_t}{P_t} - \theta_R \left(\frac{P_t(i)}{\bar{\Pi} P_{t-1}(i)} - 1 \right) \frac{Y_t}{\bar{\Pi} P_{t-1}(i)} \\ + \beta \mathbb{E}_t \frac{\lambda_{t+1}}{\lambda_t} \theta_R \left(\frac{P_{t+1}(i)}{\bar{\Pi} P_t(i)} - 1 \right) \frac{P_{t+1}}{\bar{\Pi} P_t^2} Y_{t+1} \stackrel{!}{=} 0. \end{aligned} \quad (\text{A.23})$$

Since all final good producers face identical marginal costs, all charge the same price. Therefore the i index can be dropped and one obtains the New Keynesian Phillips Curve:

$$\theta_R \left(\frac{\Pi_t}{\bar{\Pi}} - 1 \right) \frac{\Pi_t}{\bar{\Pi}} = \mathbb{E}_t \Lambda_{t,t+1} \theta_R \left(\frac{\Pi_{t+1}}{\bar{\Pi}} - 1 \right) \frac{\Pi_{t+1}}{\bar{\Pi}} \frac{Y_{t+1}}{Y_t} + mc_t \mu_t + (1 - \mu_t) \quad (\text{A.24})$$

Symmetry and the fact that there is a unit continuum of final good producers and intermediate good producers also implies that $Y_{it} = Y_t = \theta_t K_{t-1}^\alpha H_t^{1-\alpha}$.

A.4 Capital Producers

Aggregate capital in the economy is the sum of entrepreneurial and household capital:

$$K_t = K_t^E + K_t^H. \quad (\text{A.25})$$

Perfectly competitive capital producers produce new capital by purchasing the final output good (this investment is denoted I) and combining it with undepreciated capital from last period, according to the following law of motion:

$$K_t = I_t + (1 - \delta) K_{t-1} + \mathcal{C} \left(\frac{I_t}{K_{t-1}} \right) K_{t-1}. \quad (\text{A.26})$$

They face investment adjustment costs as in [Christiano et al. \(2005\)](#). Their period profits (in real terms) are:

$$\Omega_t^C = Q_t K_t - I_t - Q_t (1 - \delta) K_{t-1} \quad (\text{A.27})$$

$$= (Q_t - 1) I_t + \mathcal{C} \left(\frac{I_t}{K_{t-1}} \right) K_{t-1}. \quad (\text{A.28})$$

Since they discount real payoffs using the household's real discount factor $\beta^t \lambda_t$, their maximization problem is:

$$\max_{I_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_t \Omega_t^C. \quad (\text{A.29})$$

And the first order condition simplifies to:

$$Q_t = 1 + \mathcal{C}' \left(\frac{I_t}{K_{t-1}} \right). \quad (\text{A.30})$$

A.5 Market Clearing

Good market clearing implies that:

$$Y_t = C_t + I_t + G_t + \frac{\theta_R}{2} \left(\frac{\Pi_t}{\bar{\Pi}} - 1 \right)^2 Y_t + \frac{\theta_W}{2} \left(\frac{\Pi_t^W}{\bar{\Pi}} - 1 \right)^2 w_t + \Sigma_{ft} + \Sigma_{bt} + \frac{\kappa_H}{2} (K_t^H)^2 + c_f, \quad (\text{A.31})$$

where Σ_{ft} and Σ_{bt} are default costs of entrepreneurial firms and banks, respectively. Real repossession costs of defaulting firms are given by

$$\Sigma_{ft} = \delta_M [Q_t(1 - \delta)K_{t-1}^E + z_t K_{t-1}^E] \int_0^\infty \int_0^{\bar{\omega}_{Ft}(\omega_k)} \omega_k \omega_j dF_{jt}(\omega_j) dF_{kt}(\omega_k), \quad (\text{A.32})$$

while the real repossession costs of defaulting banks are given by

$$\Sigma_{bt} = \delta_B \left[\int_0^{\bar{\omega}_{kt}} \tilde{R}_t^L(\omega_k) L_{t-1} dF_{kt}(\omega_k) \right]. \quad (\text{A.33})$$

A.6 Proofs: Bank's Optimality Conditions

Deposits. Using the balance sheet constraint (26), deposit demand (27) and the default cutoff (23) in the bank's objective function (25) yields:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R L_{kt} + \int_{\bar{\omega}_{bkt+1}}^\infty \left((\tilde{R}_{Lkt+1}(\omega) - R_t) L_{kt} - (R_{Dkt} - R_t) \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t + (R_{t+1}^S - R_t) S_t^L + R_t \bar{B}_{kt} - c_f \right) dF_{kt+1}(\omega) \right]. \quad (\text{A.34})$$

By the Leibniz rule, the derivative with respect to the bank's deposit rate R_{Dkt} is:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} [1 - F_{kt+1}(\bar{\omega}_{bkt+1})] \left(- \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t + \epsilon_D \frac{1}{R_{Dt}} (R_{Dkt} - R_t) \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D - 1} D_t \right) = 0. \quad (\text{A.35})$$

The term $\frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} [1 - F_{kt+1}(\bar{\omega}_{bkt+1})]$ can never be zero (unless the bank fails with certainty), hence the optimal deposit rate is characterized by:

$$-\left(\frac{R_{Dkt}}{R_{Dt}}\right)^{-\epsilon_D} D_t + \epsilon_D \frac{1}{R_{Dt}} (R_{Dkt} - R_t) \left(\frac{R_{Dkt}}{R_{Dt}}\right)^{-\epsilon_D - 1} D_t = 0. \quad (\text{A.36})$$

From symmetry, it follows that: $R_{Dkt} = R_{Dt}$. Using this condition, the optimal deposit rate is given by (30). $\square$

Loan Supply. It has just been shown that the optimal choice for the deposit rate R_{Dt}^* is independent of L_{kt} . Using this result, it is shown below that the bank's objective is convex in the loan volume for any given level of long-maturity bonds S_t^L (including the optimal level). Using the optimal deposit rate (30), the balance sheet constraint (26) and the default cutoff (23) in the bank's objective function (25), the latter can be stated as a function of the loan volume for any given level of S_t^L :

$$f(L_{kt}) = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} [c_R L_{kt} + \int_{\bar{\omega}_{bkt+1}}^{\infty} \left((\tilde{R}_{Lkt+1}(\omega) - R_t) L_{kt} - (R_{Dt}^* - R_t) D_t + (R_{t+1}^S - R_t) S_t^L + R_t \bar{B}_{kt} - c_f \right) dF_{kt+1}(\omega)]. \quad (\text{A.37})$$

By the Leibniz rule, we have:

$$f'(L_{kt}) = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R + \int_{\bar{\omega}_{bkt+1}}^{\infty} \left[\tilde{R}_{Lkt+1}(\omega) - R_t \right] dF_{kt+1}(\omega) \right], \quad (\text{A.38})$$

and, as long as the bank's default threshold $\bar{\omega}_{bkt+1}$ exists (i.e. for a given amount of lending, there are states of the world in which the bank fails and other states of the world in which it is solvent):

$$f''(L_{kt}) = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[-\frac{\partial \bar{\omega}_{bkt+1}}{\partial L_{kt}} \left[\tilde{R}_{Lkt+1}(\bar{\omega}_{bkt+1}) - R_t \right] f_{kt+1}(\bar{\omega}_{bkt+1}) \right], \quad (\text{A.39})$$

where by definition of the default threshold ω_b and the implicit function theorem:

$$\frac{\partial \bar{\omega}_{bkt+1}}{\partial L_{kt}} = -\frac{\tilde{R}_L(\bar{\omega}_{bkt+1}) - R_t}{\frac{\partial \tilde{R}_L(\bar{\omega}_{bkt+1})}{\partial \omega_{bkt+1}} L_{kt}}. \quad (\text{A.40})$$

Before turning to the case in which the bank's default threshold exists (as will be the case in equilibrium), let us briefly discuss the case in which this threshold does not exist. First, the threshold may not exist because the bank makes losses from non-lending activities that exceed (with certainty) any possible payoff from lending (for a given loan rate, which is taken as given) for all levels of lending consistent with the capital requirement, i.e. $\forall L_{kt} \in [0, \bar{L}_{kt}]$. In this case, the bank fails with certainty and is obviously indifferent between any level of lending $L_{kt} \in [0, \bar{L}_{kt}]$. Second, the threshold may not exist because the payoff from non-lending activities is so high that it always exceeds any possible losses from lending $\forall L_{kt} \in [0, \bar{L}_{kt}]$. In this case, the lower bound of the integral in Eq. (A.38) becomes zero. The bank is then indifferent between any $L_{kt} \in [0, \bar{L}_{kt}]$ if:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R + \int_0^\infty \left[\tilde{R}_{L_{kt}+1}(\omega) - R_t \right] dF_{kt+1}(\omega) \right] = 0. \quad (\text{A.41})$$

and strictly prefers lending either $\bar{L}_{kt}$ (when the LHS of (A.41) is above zero) or 0 (when the LHS of (A.41) is below zero).

We now turn to the case in which the bank's default threshold exists. To shorten notation, let the realized return on physical capital for entrepreneurs (net of shocks ω) be denoted by R_{Kt} :

$$R_{Kt} = \frac{\Pi_t [Q_t(1 - \delta) + z_t]}{Q_{t-1}}, \quad (\text{A.42})$$

and let the leverage of entrepreneurial firms be denoted:

$$\Theta_t = \frac{E_t}{Q_t K_t^E}. \quad (\text{A.43})$$

The bank does not internalize any effects of changes in the bank's default cutoff on the return on physical capital or firm leverage. Then by definition of the ex-post return on loans, we

have:

$$\begin{aligned}
\frac{\partial \tilde{R}_L(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} &= -\frac{\partial \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} R_{Lkt} f_{jt+1}(\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})) \\
&+ (1 - \mu_F) \frac{R_{Kt+1}}{1 - \Theta_t} \int_0^{\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})} \omega dF_{jt+1}(\omega) \\
&+ (1 - \mu_F) \bar{\omega}_{bkt+1} \frac{R_{Kt+1}}{1 - \Theta_t} \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1}) f_{jt+1}(\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})) \tag{A.44} \\
&= [R_{Lkt} - (1 - \mu_F) R_{Lkt}] \left(-\frac{\partial \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} \right) + (1 - \mu_F) \frac{R_{Kt+1}}{1 - \Theta_t} \int_0^{\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})} \omega dF_{jt+1}(\omega) > 0. \tag{A.45}
\end{aligned}$$

The second equation follows since by definition of $\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})$:

$$\frac{\partial \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} = -\frac{\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\bar{\omega}_{bkt+1}} < 0. \tag{A.46}$$

Hence $f''(L_{kt})$ simplifies to:

$$f''(L_{kt}) = \mathbb{E}_t \Lambda_{t,t+1} \frac{(\tilde{R}_L(\bar{\omega}_{bkt+1}) - R_t)^2}{\frac{\partial \tilde{R}_L(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} L_t} f_{kt+1}(\bar{\omega}_{bkt+1}) \geq 0. \tag{A.47}$$

Summarizing the above discussion, it follows that:

$$f''(L_{kt}) \begin{cases} = 0 & \text{if } \nexists \bar{\omega}_{bkt+1} \text{ or } \tilde{R}_L(\bar{\omega}_{bkt+1}) = R_t, \\ > 0 & \text{else.} \end{cases} \tag{A.48}$$

The objective function is thus convex in L_{kt} (but not strictly convex). Hence, when $f(L)$ is linear in $[0, \bar{L}_{kt}]$, the optimal lending decision is determined by the sign of $f'(L)$, which depends on the loan rate. Note that this implies that any equilibrium with $\tilde{R}_L(\bar{\omega}_{bkt+1}) = R_t$ involves $f'(L) > 0$ and hence $L_{kt} = \bar{L}_{kt}$. On the other hand, whenever $f(L)$ has a strictly convex region in $[0, \bar{L}_{kt}]$, the Karush-Kuhn-Tucker conditions characterize a *minimum* for a given loan rate.²⁸ Hence, the optimal loan supply follows a corner solution and either $L_{kt} = 0$ or $L_{kt} = \bar{L}_{kt}$ depending on the loan rate. $\square$

²⁸To see this, note that the constraint $\gamma L_{kt} \leq \bar{L}_{kt}$ – where $\bar{L}_{kt} = \frac{\bar{B}_t}{\gamma}$ under fair-value capital requirements and $\bar{L}_{kt} = \frac{B_t^{AC}}{\gamma}$ under amortized-cost capital requirements – is clearly linear in L_{kt} .

A.7 Contracting Problem Between Banks & Entrepreneurial Firms

Let λ_t^F denote the multiplier on the firm's financing constraint Eq. (7) and λ_t^{PC} the multiplier on the bank's participation constraint Eq. (36). The first order conditions of the contracting problem are given by:

$$(K_t^E) : \mathbb{E}_t \Lambda_{t,t+1}^E \int_0^\infty \int_{\bar{\omega}_{Ft+1}(\omega_k)}^\infty \omega_j \omega_k [Q_{t+1}(1-\delta) + z_{t+1}] dF_k(\omega_j) dF_j(\omega_j) - Q_t \lambda_t^F - \lambda_t^{PC} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\int_{\bar{\omega}_{bt+1}}^\infty \frac{\partial \tilde{R}_{Lt+1}(\omega)}{\partial K_t^E} dF_k(\omega) \right] = 0 \quad (\text{A.49})$$

$$(L_{jt}) : \mathbb{E}_t \frac{\Lambda_{t,t+1}^E}{\Pi_{t+1}} \int_0^\infty \int_{\bar{\omega}_{Ft+1}(\omega_k)}^\infty (-R_{Lt}) dF_k(\omega_j) dF_j(\omega_j) + \lambda_t^F - \lambda_t^{PC} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\int_{\bar{\omega}_{bt+1}}^\infty \frac{\partial \tilde{R}_{Lt+1}(\omega)}{\partial L_t} dF_k(\omega) \right] = 0 \quad (\text{A.50})$$

$$(R_{Lt}) : \mathbb{E}_t \frac{\Lambda_{t,t+1}^E}{\Pi_{t+1}} \int_0^\infty \int_{\bar{\omega}_{Ft+1}(\omega_k)}^\infty (-L_t) dF_k(\omega_j) dF_j(\omega_j) - \lambda_t^{PC} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\int_{\bar{\omega}_{bt+1}}^\infty \frac{\partial \tilde{R}_{Lt+1}(\omega)}{\partial R_{Lt}} dF_k(\omega) \right] = 0 \quad (\text{A.51})$$

Using that $\bar{\omega}_{Ft+1}(\omega_k) = \frac{R_{Lt} L_j}{\omega_k \Pi_{t+1} [Q_{t+1}(1-\delta) K_{Et} + z_{t+1} K_{Et}]}$, these derivatives are given by:

$$\begin{aligned} \frac{\partial \tilde{R}_{Lt+1}(\omega_k)}{\partial K_{Et}} &= \frac{\omega_k (1-\delta_M) \Pi_{t+1} [Q_{t+1}(1-\delta) + z_{t+1}]}{L_t} \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \\ &+ \delta_M \frac{R_{Lt}}{K_{Et}} \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\ln(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \end{aligned} \quad (\text{A.52})$$

$$\begin{aligned} \frac{\partial \tilde{R}_{Lt+1}(\omega_k)}{\partial L_t} &= - \frac{\omega_k (1-\delta_M) \Pi_{t+1} [Q_{t+1}(1-\delta) + z_{t+1}] K_{Et}}{L_t^2} \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \\ &- \delta_M \frac{R_{Lt}}{L_t} \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\ln(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \end{aligned} \quad (\text{A.53})$$

$$\frac{\partial \tilde{R}_{Lt+1}(\omega_k)}{\partial R_{Lt}} = (1 - F_{kt+1}(\bar{\omega}_{Ft+1}(\omega_k))) - \delta_M \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\ln(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \quad (\text{A.54})$$

A.7.1 Taylor Approximation

The expectation of the ex-post realized loan rate and its derivatives are highly non-linear functions of ω_k . Therefore, to solve the model in Dynare at third order, it is necessary to manually compute a third order approximation.²⁹ The procedure follows Mendicino et al. (forthcoming). We need the approximation of just three terms involving ω_k :

$$A \equiv \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.55})$$

$$\frac{\partial A}{\partial \omega_k} = -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}\omega_k}, \quad (\text{A.56})$$

$$\begin{aligned} \frac{\partial^2 A}{\partial \omega_k^2} = & -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{1}{\sigma_{jt+1}\omega_k} \right)^2 \\ & + \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}\omega_k^2}. \end{aligned} \quad (\text{A.57})$$

$$B \equiv \omega_k \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.58})$$

$$\frac{\partial B}{\partial \omega_k} = -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}} + \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.59})$$

$$\frac{\partial^2 B}{\partial \omega_k^2} = -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}^2 \omega_k} \quad (\text{A.60})$$

$$- \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}\omega_k}. \quad (\text{A.61})$$

²⁹Derivatives of external functions are currently only implemented in Dynare up to second order.

$$C \equiv \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.62})$$

$$\frac{\partial C}{\partial \omega_k} = \frac{1}{\sigma_{jt+1}^2 \omega_k} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.63})$$

$$\frac{\partial^2 C}{\partial \omega_k^2} = -\frac{1}{\sigma_{jt+1}^2 \omega_k^2} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \quad (\text{A.64})$$

$$+ \frac{1}{\sigma_{jt+1}^3 \omega_k^2} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right)^2 \quad (\text{A.65})$$

$$- \frac{1}{\sigma_{jt+1}^3 \omega_k^2} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right). \quad (\text{A.66})$$

Then, as in Mendicino et al. (forthcoming), the expected ex-post realized loan rate can be approximated as:

$$\mathbb{E}_t \tilde{R}_{t+1}^L \approx \sum_{i=1}^N \left(\int_{x_i}^{x_{i+1}} T \left(\tilde{R}_{t+1}^L \right) (\omega_k) dF_{kt+1}(\omega_k) \right) + [1 - F_{kt+1}(x_{N+1})] R_t^L. \quad (\text{A.67})$$

where the Taylor Approximation of the ex-post realized loan rate around a point $\bar{x}_i = \frac{x_i + x_{i+1}}{2}$ is given by:

$$T \left(\tilde{R}_{t+1}^L \right) (\omega_k) = \tilde{R}_{t+1}^L(\bar{x}_i) + \frac{\partial \tilde{R}_{t+1}^L}{\partial \omega_k} (\omega_k - \bar{x}_i) + \frac{1}{2} \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2} (\omega_k - \bar{x}_i)^2. \quad (\text{A.68})$$

Using the expressions just derived:

$$T \left(\tilde{R}_t^L \right) (\omega_k) = \frac{(1 - \delta_M) \Pi_{t+1} [Q_{t+1}(1 - \delta) + z_{t+1}] K_t^E}{L_t} T(B) + R_t^L (1 - T(A)). \quad (\text{A.69})$$

Then:

$$\begin{aligned} \int_{x_i}^{x_{i+1}} T \left(\tilde{R}_{t+1}^L \right) (\omega_k) dF_{kt+1}(\omega_k) &= Q_0(\bar{x}_i) + Q_1(\bar{x}_i) \int_{x_i}^{x_{i+1}} \omega_k dF_{kt+1}(\omega_k) \\ &\quad + Q_2(\bar{x}_i) \int_{x_i}^{x_{i+1}} \omega_k^2 dF_{kt+1}(\omega_k), \end{aligned} \quad (\text{A.70})$$

where

$$Q_0(\bar{x}_i) = [F_{kt+1}(\omega_{i+1}) - F_{kt+1}(\omega_i)] \left[\tilde{R}_{t+1}^L(\bar{x}_k) - \bar{x}_i \frac{\partial \tilde{R}_{t+1}^L}{\partial \omega_k} + \frac{1}{2} \bar{x}_i^2 \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2} \right], \quad (\text{A.71})$$

$$Q_1(\bar{x}_i) = \left[\frac{\partial \tilde{R}_{t+1}^L}{\partial \omega_k} - \bar{x}_i \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2} \right], \quad (\text{A.72})$$

$$Q_2(\bar{x}_i) = \frac{1}{2} \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2}. \quad (\text{A.73})$$

We proceed similarly for the derivatives of the ex-post loan rate.

B Calibration

This Appendix presents additional details on the calibration strategy, and provides full details of the data sources.

where $\epsilon_{mw} \sim N(0, 1)$. The length of the simulation is 1700 quarters after a burn-in of 1500 periods. The burn-in period ensures that the ergodic distribution is reached.³⁰ We specify priors on some parameters due to SMM's known tendency to pick parameters not supported by micro data (see [An and Schorfheide, 2007](#) and [Ruge-Murcia, 2012](#)).³¹

Parameter	Description	Prior	Variance
ϕ_Π	Taylor Rule Weight Inflation	1.5	6
ϕ_Y	Taylor Rule Weight Inflation	0.5	2
ϕ_R	Taylor Rule Smoothing	1.5	2
κ	Investment Adjustment Costs	7	40

Table 4: *Priors*

B.1 Data Sources

Gross Domestic Product (GDP) deflator OECD Quarterly National Accounts. Euro Area (20 countries). Deflator, OECD reference year, seasonally adjusted. Millions of Euro. Index.

³⁰We follow [Born and Pfeifer \(2014\)](#) who use the same burn-in length.

³¹As discussed in [Born and Pfeifer \(2014\)](#), these priors are relatively flat.

Population OECD Quarterly National Accounts. Euro Area (20 countries). Total Population. Thousands.

GDP OECD Quarterly National Accounts. Euro Area (20 countries). Gross domestic product - expenditure approach. National currency, current prices, quarterly levels, seasonally adjusted. Millions of Euro. *Transformation*: Divided by GDP deflator and total population.

Consumption Private final consumption expenditure. OECD Quarterly National Accounts. Euro Area (20 countries). National currency, current prices, quarterly levels, seasonally adjusted. Millions of Euro. *Transformation*: Divided by GDP deflator and total population.

Investment OECD Quarterly National Accounts. Euro Area (20 countries). Gross fixed capital formation. National currency, current prices, quarterly levels, seasonally adjusted. Millions of Euro. *Transformation*: Divided by GDP deflator and total population.

Inflation Log-change in GDP Deflator.

Employment OECD Quarterly National Accounts. Euro Area (20 countries). Employment, total (Persons). Thousands.

Hours OECD Quarterly National Accounts. Euro Area (20 countries). Employment, total (Hours Worked). Millions. *Transformation*: Demeaned Hours/Employment ($H = 1$ in the model corresponds to average hours worked).

Wages OECD Quarterly National Accounts. Euro Area (20 countries). Wages, total. Millions. *Transformation*: Wages divided by total hours worked and GDP deflator.

Rate of return on equity (RoE), bank default probability, and firm default probability Mendicino et al. (forthcoming).

Loan-to-bond ratio [Hoffmann et al. \(2019\)](#)

Loans ECB Statistical Data Warehouse. The ECB does not report values for a hypothetical constant composition Euro Area, such that care must be taken to compute relationships with the measure of GDP for the corresponding countries. For this reason, we construct data for the following countries: Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, Luxembourg, Netherlands and Portugal.

The ECB data we use is only available monthly, so for each of the variables below, population weighted averages are computed for each quarter. Let $\omega_{ct} = \frac{Population_{ct}}{\sum_c Population_{ct}}$, where countries are indexed by c , quarters by t and month by m . ω_c are computed from the same OECD population data used above. Then the population-weighted averages for each quarter of variable x are:

$$\bar{x}_t = \frac{1}{4} \sum_{m \in t} \sum_c \omega_c x_{cm} \quad (\text{A.74})$$

(L1) : Loans vis-a-vis euro area NFCs reported by MFIs excl. ESCB. Up to 1 year maturity. Euro area (changing composition) counterpart, Non-Financial corporations (S.11) sector, denominated in Euro. Outstanding amounts at the end of the period (stocks). Monthly.

(L2) : Loans vis-a-vis euro area NFCs reported by MFIs excl. ESCB. Over 1 and up to 5 years maturity. Euro area (changing composition) counterpart, Non-Financial corporations (S.11) sector, denominated in Euro. Outstanding amounts at the end of the period (stocks). Monthly.

(L3) : Loans vis-a-vis euro area NFCs reported by MFIs excl. ESCB. Over 5 years maturity. Euro area (changing composition) counterpart, Non-Financial corporations (S.11) sector, denominated in Euro. Outstanding amounts at the end of the period (stocks). Monthly.

Sum of: (L1) + (L2) + (L3) of all countries listed above.

Transformation: Population weighted average for each quarters. Seasonally adjusted with X-13 ARIMA using the R package seasonal. Divided by nominal GDP and multiplied by real per capita GDP to get real per-capita loans. To compute the average loan-to-GDP ratio the sum of real per capita GDP for the same countries is used.

Loan rate spread ECB Statistical Data Warehouse.

(LR1): Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER), Credit and other institutions (MFI except MMFs and central banks) reporting sector - Loans, Up to 1 year original maturity, Outstanding amount business coverage, Non-Financial corporations (S.11) sector, denominated in Euro. Monthly.

(SR1): Euro Interbank Offered Rate.

(LR2): Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER), Credit and other institutions (MFI except MMFs and central banks) reporting sector - Loans, Over 1 and up to 5 years original maturity, Outstanding amount business coverage, Non-Financial corporations (S.11) sector, denominated in Euro. Monthly.

(SR2): Yield curve spot rate, 2 year maturity. Government bond, nominal, all issuers whose rating is triple A - Euro area (changing composition). Monthly.

(LR3): Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER), Credit and other institutions (MFI except MMFs and central banks) reporting sector - Loans, Over 5 years original maturity, Outstanding amount business coverage, Non-Financial corporations (S.11) sector, denominated in Euro. Monthly.

(SR3): Yield curve spot rate, 5 year maturity. Government bond, nominal, all issuers whose rating is triple A - Euro area (changing composition). Monthly.

Transformation:

$$\frac{((LR1) - (SR1))(L1) + ((LR2) - (SR2))(L2) + ((LR3) - (SR3))(L3)}{(L1) + (L2) + (L3)}$$

of all countries listed above. Compute population weighted average for each quarters.

Safe Rate ECB Statistical Data Warehouse. Using the data just described, we approximate the safe rate (the data counterpart to R_t) as:

$$\frac{(SR1)(L1) + (SR2)(L2) + (SR3)(L3)}{(L1) + (L2) + (L3)}$$

of all countries listed above. Then we compute the population weighted average for each quarter.

C Details Computation Consumption Equivalents

Denote variable x in model a by x_t^a and in model b by x_t^b , where a and b will be specified below. To compute consumption equivalents (CE) Δ_{CE} , we first calculate the expected

lifetime utility at time t under two different models:³²

$$\mathbb{E}(U_t^a) = \mathbb{E} \left(\mathbb{E}_t \sum_{n=0}^{\infty} \beta^n \left[\frac{\{C_{t+n}^a(h) - bC_{t+n-1}^a(h)\}^{1-\sigma}}{1-\sigma} - \frac{\xi_H H_{t+n}^a(h)^{1+\varphi_H}}{1+\varphi_H} \right] \right) \quad (\text{A.75})$$

and similarly for model b . Next, the expected utility in model b can be related to the utility in model a as follows:

$$\mathbb{E}(U_t^a) = \mathbb{E} \left(\mathbb{E}_t \sum_{n=0}^{\infty} \beta^n \left[\frac{\{(1 + \Delta_{CE})(C_{t+n}^b(h) - bC_{t+n-1}^b(h))\}^{1-\sigma}}{1-\sigma} - \frac{\xi_H H_{t+n}^b(h)^{1+\varphi_H}}{1+\varphi_H} \right] \right) \quad (\text{A.76})$$

Under log-utility ($\sigma = 1$), using Eq. (A.75) in Eq. (A.76) yields:

$$\mathbb{E}(U_t^a) = \frac{\ln(1 + \Delta_{CE})}{1 - \beta} + \mathbb{E}(U_t^b) \quad (\text{A.77})$$

It follows:

$$\Delta_{CE} = \exp[(1 - \beta)(\mathbb{E}\{U_t^a - U_t^b\})] - 1 \quad (\text{A.78})$$

Model a is for example of the model under fair value requirements, while model b is the model under amortized-cost requirements. The interpretation is that households, could they choose, would require $100\Delta_{CE}\%$ of consumption in every period to remain in the economy with amortized-cost value requirements.

³²This may be interpreted as the expected utility of a person born at an arbitrary time t in the respective economy.

D Additional Tables and Figures

Table 5: *Parameter Values*

Symbol	Value	Description
α	0.25	Capital Share Production
β	0.995	HH Discount Factor
δ	0.025	Capital Depreciation Rate
θ_R	55.4563	Rotemberg Price Adjustment Cost
θ_W	1157.6892	Rotemberg Wage Adjustment Cost
φ_H	1	Inverse Frisch Elasticity of Labor
μ	7.25	Elasticity of Substitution Final Goods
ϵ_W	5	Elasticity of Substitution Labor
κ	7.5732	Investment Adjustment Cost
κ_H	0.0063535	Capital Management Cost
σ	1	HH Risk Aversion (Consumption)
ξ_H	0.83033	Disutility of Labor
θ_B	0.965	Survival Bankers
θ_E	0.97048	Survival Entrepreneurs
γ	0.08	Regulatory Capital Requirement
ϕ_Π	1.0664	Taylor Rule Weight Inflation
ϕ_y	0.57192	Taylor Rule Weight Output
ϕ_R	0.57954	Taylor Rule Weight Smoothing Weight
ξ_B	0.50078	Endowment Bankers
ξ_E	3.8038e-05	Endowment Entrepreneurs
b	0.76383	Habit Consumption
δ_M	0.3	Loss Entrepreneurial Default
δ_B	0.3	Loss Bank Default
$\bar{\sigma}_{\omega_j}$	0.16947	Steady State Std Idiosyncratic Shock
$\bar{\sigma}_{\omega_k}$	0.11443	Steady State Std Island Shock
σ_{ω_j}	0.00056851	Std Idiosyncratic Risk Shock
σ_{ω_k}	0.043043	Std Island Risk Shock
ρ_{ω_j}	0.22872	Autocorr. Idiosyncratic Risk Shock
ρ_{ω_k}	0.98459	Autocorr. Island Risk Shock
ρ_θ	0.021021	Autocorr. Productivity Shock
ρ_p	0.49082	Autocorr. Cost Push Shock
ρ_g	0.8297	Autocorr. Gov. Spending Shock
ρ_τ	0.88458	Autocorr. Monetary Policy Shock
σ_θ	0.00043939	Std. Productivity Shock
σ_p	0.18936	Std. Cost Push Shock
σ_g	0.0038184	Std. Gov. Spending Shock
σ_τ	0.00095329	Std. Monetary Policy Shock
S^L	1.1424	Real Supply Central Bank Asset
m	13.6	Gov. Bond Maturity
c_f	0.0026084	Bank Fixed Cost
c_R	0.0251	Bank Relationship Lending Benefit
ϵ_D	-400	Deposits Elasticity of Substitution